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Approximate Bayesian computation (ABC) method for estimating parameters of the gamma process using noisy data

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ABSTRACT

The stochastic gamma process (GP) has emerged as a versatile model of degradation processes affecting the performance of engineering structures and components. The GP model captures the temporal uncertainty in the degradation process, whereas the Bayesian approach is used to account for uncertainty in the model parameters. Although the conceptual approach to GP parameter estimation is straightforward, its practical implementation is quite challenging. The reason is that degradation data required for parameter estimation are invariably contaminated by measurement noise, which turns the likelihood function into a high-dimensional multivariate integral. To circumvent this difficulty, the paper presents a novel and practical approach based on the approximate Bayesian computation (ABC) technique. The ABC method is a simulation-based approach that does not require an explicit formulation of the likelihood function. Instead, ABC uses an "accept-reject" mechanism to perform the posterior computation. To reduce the rejection rate, the Markov chain Monte Carlo (MCMC) approach is used within the ABC algorithm. The proposed ABC-MCMC algorithm is shown to be computationally more efficient than the traditional likelihood-based method without compromising the numerical accuracy. A case study involving the modelling of corrosion in nuclear piping systems is presented, which confirms the practical usefulness of the proposed method.

1. Introduction

1.1. Background

Statistical estimation of the stochastic gamma process (GP) parameters requires degradation measurement data over some specific time intervals. If degradation data can be measured accurately, the formulation of the likelihood function is fairly straightforward. However, in practical situations, degradation data are collected using various non-destructive inspection tools and probes, which tend to introduce random noise in measurements. A more practical case is that in which the degradation processes are measured at several different locations and inspection intervals with a noisy probe; such a sample of multiple sets of noisy measurements poses a great deal of difficulty in solving the parameter estimation problem by a likelihood-based method. The reason for the difficulty is that the presence of measurement error introduces a dependence between the measured increments of degradation across different times of inspection. As shown by Lu [1] and Yuan [2], the degradation sample likelihood of the GP model involves a multivariate likelihood function which is not easy to evaluate in the process of parameter estimation.

Besides measurement noise, lack of inspection data is another challenge to the parameter estimation problem in modelling any degradation process. As an example, degradation data collected from a nuclear reactor tends to be quite small in order to limit the exposure of inspectors to the intense radiation field that surrounds the reactor components. Therefore, given small samples of noisy measurements of a degradation process, it is important to account for the parameter uncertainty in the parameter estimation problem.

A Bayesian framework for the GP model is well-suited to quantify the associated measurement and sampling uncertainties. However, the case of noisy measurement data makes it difficult to solve the two key steps of the Bayesian inference, namely: (1) likelihood evaluation and (2) computing the normalizing constant. Although the likelihood-based MCMC method is commonly used to circumvent the latter step, i.e., the computation of the normalizing constant, this approach does not solve the problem associated with the complexity of the likelihood function. For example, Son et al. [3] implemented Gibbs sampler for stochastic filtering of the hidden degradation states of a noisy gamma degradation process. In their study, the conditional densities of the hidden degradation states, which comprise a large number of integrals, show the computational complexity of the problem. To reduce the complexity,

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they used marginal conditional densities for carrying out the state estimation procedure. Similarly, Qin et al. [4] implemented MCMC with the data augmentation technique to deal with the computational complexity related to posterior computation of their proposed model parameters from noisy oil and gas pipeline data. In summary, the likelihood-based Bayesian inference of GP parameters from noisy data is a computationally challenging task.

The approximate Bayesian computation (ABC) method has emerged as a way to provide likelihood-free solutions of the Bayesian inference problems involving the likelihood functions that are either analytically intractable or computationally expensive to evaluate. The ABC method uses the Monte Carlo method to simulate data directly from a forward model and retain relevant samples based on a suitable distance function in a way similar to that of the "acceptance-rejection" method. The basic premise of the ABC method is that if data simulation is computationally cheap, then computational burden associated with a complex likelihood-based Bayesian inference can be significantly reduced. Thus, the main objective of this study is to present a likelihood-free ABC method for the Bayesian inference of GP parameters based on noisy data obtained from multiple repeat measurements of a degradation process.

1.2. Related literature

1.2.1. Application of GP in degradation modelling

The stochastic GP has been applied to the modelling of a wide variety of degradation processes affecting engineering structures and components, such as corrosion [4–7], creep [8], crack growth [9–11], and chloride attack in concrete structures [12]. Numerous recent studies in reliability engineering used the GP process as a primary degradation model. Mireh et al. [13] used a copula-based approach to perform reliability analysis for systems within the GP degradation framework. The authors showed that when dependency in the degradation process exists, not considering it will produce erroneous reliability estimates. Similarly, Wang et al. [14] designed experiments to perform reliability analysis of a cantilever beam under cyclic loading with a GP deterioration model. Liu et al. [15] proposed a method for reliability analysis of systems having monotonic degradation based on Bayesian model averaging. They used the GP and inverse Gaussian process as the candidate models for the system degradation. Giorgio et al. [16] used the transformed GP to model degradation growth of a cargo ship engine. They used the Markov chain Monte Carlo (MCMC) technique to estimate the model parameters, which are subsequently used for estimating residual reliability and lifetime. Guida et al. [17] implemented the MCMC method to estimate the parameters of the non-homogeneous GP used to model a sliding metal wear degradation phenomenon.

1.2.2. Development of ABC and its applications in engineering

The basic idea of the likelihood-free ABC algorithm was first described by Rubin [18] in 1984, where this method was presented as an intuitive way to understand the Bayesian posterior computation by simulation [19]. Later Tavaré et al. [20] used the ABC algorithm as an "accept-reject" method to infer the genealogy of DNA sequence data in a discrete sample space [19,21]. Pritchard et al. [22] first extended the ABC method to a continuous sample space to solve population genetics problems. Finally, Beaumont et al. [23] coined the term "approximate Bayesian computation" in their work related to population genetics. During the last two decades, development and application of more sophisticated ABC algorithms in the field of biological sciences [24–31] clearly demonstrate its efficiency and usefulness in Bayesian computation. Plenty of review papers on ABC [19,21,32–35] are also available in the literature.

Application of ABC in the engineering field is very limited [1,36–40]. Abdessalem et al. [36] implemented the likelihood-free ABC method for model selection and parameter estimation in non-linear system identification problems. They showed that the performance of ABC is quite promising for systems with complex behaviours. Later they improved the efficiency of the method by using ABC in conjunction with ellipsoidal nested sampling technique [37]. Chen et al. [41] implemented ABC with the population Monte Carlo sampling technique to approximate the lifetime distributions of the electromagnetic relay components in United States military aircrafts.

1.3. Objectives and organization

The specific objectives of this study are as follows. Firstly, to present a Bayesian framework for the estimation of the GP parameters from noisy data. Secondly, to understand the theory behind likelihood-free inference and develop a strategy for implementation of an efficient version of ABC, called ABC using MCMC (ABC-MCMC) [27], for the estimation purpose. Thirdly, to compare the proposed method in terms of accuracy and computing time with a traditional Bayesian inference technique, the Metropolis–Hastings (MH) algorithm [42], through simulation-based examples. Finally, to demonstrate the applicability of the proposed method by solving a practical problem based on degradation data from a Canadian nuclear reactor. The MH algorithm is a very popular likelihood-based MCMC (L-MCMC) technique for Bayesian computation [43,44]. Note that this paper uses the term L-MCMC to indicate the likelihood-based MH algorithm.

The paper starts with a description of the stochastic GP model along with a likelihood-based Bayesian parameter estimation framework in Section 2. Section 3 presents a comprehensive review of the theory behind likelihood-free inference and the related algorithms. Section 4 is dedicated to the numerical examples to investigate the efficiency and applicability of the proposed method. Finally, Section 5 provides the conclusions of the study.

2. Problem formulation

2.1. Degradation model

Under the assumption of an additive noise model, the measured degradation $Y(t)$ is represented as a sum of random initial degradation A, true degradation $X(t)$, and measurement error or noise Z, which is a time invariant random variable [45]:

$$Y(t) = A + X(t) + Z \tag{1}$$

Thus, a noisy degradation measurement of component i at time j is represented by $y_{ij} = a_i + x_{ij} + z_{ij}$, where a_i ($y_{i0} = a_i$) is the related initial degradation, x_{ij} is the true degradation growth, and z_{ij} ($z_{i0} = 0$) is the random sizing error. The unknown initial degradation a_i (constant for each component but variable over the entire population) is a realization of A with PDF $f_A(\bullet) = \mathcal{N}(\mu_A, \sigma_A)$, where $\mathcal{N}(\bullet,\bullet)$ represents a normal density function, and μ_A and σ_A are the mean and standard deviation (SD) of A, respectively. The true degradation x_{ij} is a realization of the degradation process $X(t)$ which is a stochastic GP. The sizing error z_{ij} is a realization of the independent and identically distributed (IID) random variable Z with the PDF $f_Z(\bullet) = \mathcal{N}(0, \sigma_Z)$, where σ_Z is the SD of the measurement noise.

A general monotonic degradation process can be modelled as a non-stationary GP, $X(t)$, $t \geq 0$, with a non-linear, time dependent shape function $a(t)$, and scale parameter β [46]. For $0 \leq t_1 \leq t_2 \leq \cdots \leq t_n$, increments of $X(t)$, i.e., $X(t_1) - X(0), X(t_2) - X(t_1), \cdots, X(t_n) - X(t_{n-1})$, are independent gamma distributed random variables. An increment $\Delta X = X(t + \Delta t) - X(t)$ for $t \geq 0$ has the following gamma probability density function (PDF):

$$\mathcal{G}[\Delta x | a(t + \Delta t) - a(t), \beta] = \frac{(\Delta x/\beta)^{a(t+\Delta t)-a(t)-1}}{\beta\, \Gamma(a(t + \Delta t) - a(t))} \exp(-\Delta x/\beta)\, \mathbb{I}_{[0,+\infty)}(\Delta x) \tag{2}$$

where $\Gamma(s) = \int_0^{+\infty} t^{s-1} e^{-t} dt$ is the complete gamma function and $\mathbb{I}_{[0,+\infty)}(\Delta x) = 1$, if $\Delta x \in [0, +\infty)$; otherwise it is 0.

The shape parameter $a(t)$ for $t \geq 0$ is a non-decreasing, right-continuous, real-valued function such as a commonly used power law function, $a(t) = \alpha t^{\eta}$, where $\alpha > 0$ and $\eta > 0$ are constants and $a(t = 0) = 0$ [46]. The case of $\eta = 1$ represents a stationary GP. The expectation, variance and coefficient of variation (COV) of $X(t)$ are given as

$$\mathbb{E}[X(t)] = \alpha t^{\eta}\beta,\ \ \mathrm{Var}[X(t)] = \alpha t^{\eta}\beta^2 \text{ and } \mathrm{COV}[X(t)] = \frac{1}{\sqrt{\alpha t^{\eta}}} \tag{3}$$

The likelihood function for the GP parameters, α, η, and β, along with other distribution parameters, μ_A, σ_A, and σ_Z, given measured degradation data collected from N components can be written as

$$\begin{aligned}&\mathcal{L}(\alpha, \eta, \beta, \mu_A, \sigma_A, \sigma_Z|\Delta\mathbf{y}_1, \Delta\mathbf{y}_2, \cdots, \Delta\mathbf{y}_N)\\ &= \prod_{i=1}^{N}\int_{a_i}\int_{\Delta\mathbf{z}_i} f_{\Delta\mathbf{X}_i}(\Delta\mathbf{y}_i - \Delta\mathbf{z}_i|a_i) f_A(a_i) f_{\Delta\mathbf{Z}_i}(\Delta\mathbf{z}_i) da_i d(\Delta\mathbf{z}_i)\end{aligned} \tag{4}$$

where $\Delta\mathbf{y}_i$ and $\Delta\mathbf{z}_i$ are measurement and noise vectors, respectively; and $f_{\Delta\mathbf{X}_i}(\bullet)$ is the joint distribution of the true degradation growths. The likelihood function in Eq. (4) is a multi-dimensional integration that has a dimension of $(N + \sum_{i=1}^{N} m_i)$, where m_i is the number of component-specific measurements. The derivation of the likelihood function is presented in Appendix A. For convenience, the vector of the model parameters is denoted as $\mathbf{\Theta} = \{\alpha, \eta, \beta, \mu_A, \sigma_A, \sigma_Z\}^T$ and observed data as $\mathbf{D}_{obs} = \{\Delta\mathbf{y}_1, \Delta\mathbf{y}_2, \cdots, \Delta\mathbf{y}_N\}$.

2.2. Likelihood-based Bayesian inference

The Bayesian inference begins with assigning a prior distribution $f(\mathbf{\Theta})$ to the parameter vector $\mathbf{\Theta}$ of the degradation process. Given a sample of observation data $\mathbf{D}_{obs}$, the posterior distribution of $\mathbf{\Theta}$ is obtained as

$$f(\mathbf{\Theta}|\mathbf{D}_{obs}) = \frac{\mathcal{L}(\mathbf{\Theta}|\mathbf{D}_{obs}) f(\mathbf{\Theta})}{\int_{\mathbf{\Theta}} \mathcal{L}(\mathbf{\Theta}|\mathbf{D}_{obs}) f(\mathbf{\Theta}) d\mathbf{\Theta}} \tag{5}$$

using the likelihood function $\mathcal{L}(\mathbf{\Theta}|\mathbf{D}_{obs})$ and the normalizing constant, which is the integral term in the denominator.

The evaluation of the normalizing constant is a computationally intensive task because it involves a multivariate integration over the parameter space. As a result, direct evaluation of the posterior distribution is almost impossible in practical situations. To avoid the computation of the normalizing constant, the L-MCMC method (i.e., the likelihood-based MH sampler) can be used.

The L-MCMC sampler defines an acceptance probability $\mathcal{A}(\mathbf{\Theta}^{(k)} \to \mathbf{\Theta}^*)$ for the current sample $\mathbf{\Theta}^*$ given the previous sample $\mathbf{\Theta}^{(k)}$ based on the posterior distribution and a chosen proposal distribution $q(\mathbf{\Theta}^{(k)} \to \mathbf{\Theta}^*)$, i.e.,

$$\mathcal{A}(\mathbf{\Theta}^{(k)} \to \mathbf{\Theta}^*) = \min\left\{1, \frac{f(\mathbf{\Theta}^*|\mathbf{D}_{obs}) q(\mathbf{\Theta}^* \to \mathbf{\Theta}^{(k)})}{f(\mathbf{\Theta}^{(k)}|\mathbf{D}_{obs}) q(\mathbf{\Theta}^{(k)} \to \mathbf{\Theta}^*)}\right\} \tag{6}$$

After substituting the posterior distribution from Eq. (5), the acceptance probability can be written as

$$\mathcal{A}(\mathbf{\Theta}^{(k)} \to \mathbf{\Theta}^*) = \min\left\{1, \frac{\mathcal{L}(\mathbf{\Theta}^*|\mathbf{D}_{obs}) f(\mathbf{\Theta}^*) q(\mathbf{\Theta}^* \to \mathbf{\Theta}^{(k)})}{\mathcal{L}(\mathbf{\Theta}^{(k)}|\mathbf{D}_{obs}) f(\mathbf{\Theta}^{(k)}) q(\mathbf{\Theta}^{(k)} \to \mathbf{\Theta}^*)}\right\} \tag{7}$$

It is clear that the acceptance probability can be calculated only by evaluating the likelihood function since the normalizing constant gets canceled. The computation of the likelihood function plays a key role in the algorithm which generates a Markov chain that has a stationary distribution of $f(\mathbf{\Theta}|\mathbf{D}_{obs})$. The steps to generate L-MCMC samples $\mathbf{\Theta}^{(k)} \sim f(\mathbf{\Theta}|\mathbf{D}_{obs})$, $k = 1, 2, \cdots, n$, are given in Algorithm 1. For a more detailed discussion on the method, the reader is referred to [42].

2.3. Remarks

The likelihood function $\mathcal{L}(\mathbf{\Theta}|\mathbf{D}_{obs})$ given in Eq. (4) and the normalizing constant $[\int_{\mathbf{\Theta}} \mathcal{L}(\mathbf{\Theta}|\mathbf{D}_{obs}) f(\mathbf{\Theta}) d\mathbf{\Theta}]^{-1}$ in Eq. (5) contain complex high-dimensional integrals. Consequently, numerical evaluation of these integrals multiple-times is computationally challenging. Although the L-MCMC method helps by eliminating the need to compute the normalizing constant, likelihood evaluation remains a numerical challenge. The likelihood evaluations in L-MCMC are computationally demanding and generally performed using the Monte Carlo simulation method. This computational complexity can be avoided to some extent by implementing an efficient quasi-Monte Carlo-based integration scheme proposed by Lu et al. [6]. However, the next section shows that the use of the ABC method completely avoids the evaluation of both the normalizing constant and the likelihood function, thereby providing an efficient way of performing Bayesian inference on the stochastic GP model.

3. Approximate Bayesian computation (ABC)

The ABC algorithm approximates the posterior distribution $f(\mathbf{\Theta}|\mathbf{D}_{obs}) \propto \mathcal{L}(\mathbf{\Theta}|\mathbf{D}_{obs}) f(\mathbf{\Theta})$ by generating samples from a joint distribution of the parameter vector $\mathbf{\Theta}$ and simulated data $\mathbf{D}$ using forward modelling and an accept-reject mechanism (the forward model in this study is GP and $\mathbf{D}$ can be obtained by simulating GP sample paths for given values of $\mathbf{\Theta}$). To avoid the likelihood evaluation, the method first generates a set of candidate parameter vectors $\{\mathbf{\Theta}^{(1)}, \mathbf{\Theta}^{(2)}, \cdots\}$ by sampling from the prior density $f(\mathbf{\Theta})$; then simulates the corresponding set of complete data sets $\{\mathbf{D}^{(1)}, \mathbf{D}^{(2)}, \cdots\}$ from a forward model $\mathcal{M}(\mathbf{D}|\mathbf{\Theta})$; and finally accepts those $\{\mathbf{\Theta}^{(k)}, \mathbf{D}^{(k)}\}$ combinations which satisfy the condition $\rho(\mathbf{D}, \mathbf{D}_{obs}) \leq \epsilon$, where $\rho(\mathbf{D}, \mathbf{D}_{obs})$ is a distance function and $\epsilon \geq 0$ is a tolerance threshold. A conceptual overview of likelihood-free Bayesian inference is presented in Fig. 1.

The final outcome of this algorithm are samples of parameter vectors and simulated data from the distribution $f(\mathbf{\Theta}, \mathbf{D}|\rho(\mathbf{D}, \mathbf{D}_{obs}) \leq \epsilon)$. Hence, the true posterior distribution, approximated using ABC, can be written as

$$f(\mathbf{\Theta}|\mathbf{D}_{obs}) \approx f_{\epsilon}(\mathbf{\Theta}|\mathbf{D}_{obs}) = \int_{\mathbf{D}} f(\mathbf{\Theta}, \mathbf{D}|\rho(\mathbf{D}, \mathbf{D}_{obs}) \leq \epsilon) d\mathbf{D} \tag{8}$$

where $f_{\epsilon}(\mathbf{\Theta}|\mathbf{D}_{obs})$ is the marginal ABC posterior of $\mathbf{\Theta}$. For a detailed discussion on the convergence of the ABC posterior, refer to Appendix B.

A good approximation of the posterior can be achieved through the selection of a sufficiently small ϵ value. However, the tolerance threshold ϵ is selected based on a trade-off between the computational cost and the accuracy of the algorithm. In other words, a smaller ϵ will provide higher accuracy to the algorithm with a higher computational cost.

Following the intuitive idea of ABC, Pritchard et al. [22] proposed the first basic ABC algorithm, also known as the ABC rejection sampler. The steps of the sampler are presented in Algorithm 2. A major drawback of the ABC rejection sampler is that the acceptance rate is very low when the prior consists of a considerably wider support. To circumvent this problem, an advanced hybrid ABC algorithm, ABC-MCMC, is introduced next.

3.1. ABC-MCMC sampler

Marjoram et al. [27] introduced a likelihood-free MCMC sampler based on the ABC method. The ABC-MCMC sampler draws samples directly from the ABC posterior $f_{\epsilon}(\mathbf{\Theta}|\mathbf{D}_{obs})$ instead of the true posterior $f(\mathbf{\Theta}|\mathbf{D}_{obs})$. The necessary steps for the implementation of ABC-MCMC are shown in Algorithm 3. The authors proved the convergence of the sampler directly by satisfying the detailed balance equation. However, it can be shown that ABC-MCMC is, in fact, a special case of the

Initialize $\boldsymbol{\Theta}^{(k)} \sim f(\boldsymbol{\Theta})$; set $k = 1$
for $k = 1$ to $(n - 1)$ **do**
  Generate $\boldsymbol{\Theta}^*$ from the proposal distribution $q(\boldsymbol{\Theta}^{(k)} \to \boldsymbol{\Theta}^*)$
  Calculate the acceptance probability

$$\mathcal{A}(\boldsymbol{\Theta}^{(k)} \to \boldsymbol{\Theta}^*) = \min\left\{1, \frac{\mathcal{L}(\boldsymbol{\Theta}^* \mid \mathbf{D}_{obs}) f(\boldsymbol{\Theta}^*) q(\boldsymbol{\Theta}^* \to \boldsymbol{\Theta}^{(k)})}{\mathcal{L}(\boldsymbol{\Theta}^{(k)} \mid \mathbf{D}_{obs}) f(\boldsymbol{\Theta}^{(k)}) q(\boldsymbol{\Theta}^{(k)} \to \boldsymbol{\Theta}^*)}\right\}$$

  Simulate u from the uniform distribution $\mathcal{U}[0, 1]$
  Set

$$\boldsymbol{\Theta}^{(k+1)} = \begin{cases} \boldsymbol{\Theta}^*, & \text{if } u \le \mathcal{A}(\boldsymbol{\Theta}^{(k)} \to \boldsymbol{\Theta}^*) \\ \boldsymbol{\Theta}^{(k)}, & \text{otherwise} \end{cases}$$

end for

Algorithm 1. Likelihood-based MH sampler (L-MCMC).

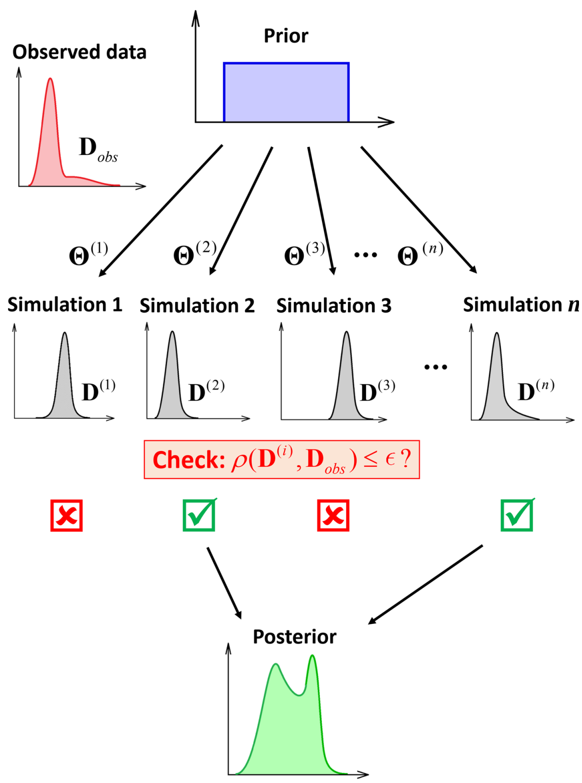

Fig. 1. A conceptual overview of the ABC method (adapted from [21]).

for $k = 1$ to n **do**
  repeat
    Generate $\boldsymbol{\Theta}^*$ from the prior $f(\boldsymbol{\Theta})$
    Simulate a data set $\mathbf{D}^*$ from $\mathcal{M}(\mathbf{D} \mid \boldsymbol{\Theta}^*)$
    Accept $\boldsymbol{\Theta}^*$, if $\rho(\mathbf{D}^*, \mathbf{D}_{obs}) \le \epsilon$
  until Acceptance
  Set $\boldsymbol{\Theta}^{(k)} = \boldsymbol{\Theta}^*$
end for

Algorithm 2. ABC rejection sampler.

likelihood-based MH sampler (L-MCMC), where the likelihood-evaluation step can be avoided by dividing the acceptance criteria into a few independent simulation steps. The derivation of the ABC-MCMC method is presented in Appendix C.

The acceptance probability in ABC-MCMC neither involves the likelihood function, nor the normalizing constant, which makes the sampler highly efficient. Note that if the proposal distribution is symmetric, then the acceptance probability depends only on the prior distribution:

$$\mathcal{A}(\boldsymbol{\Theta} \to \boldsymbol{\Theta}^*) = \min\left\{1, \frac{f(\boldsymbol{\Theta}^*)}{f(\boldsymbol{\Theta})}\right\}, \quad \text{if } q(\boldsymbol{\Theta} \to \boldsymbol{\Theta}^*) = q(\boldsymbol{\Theta}^* \to \boldsymbol{\Theta}) \tag{9}$$

Additionally, if the prior is uniform, then the acceptance probability becomes equal to 1, which implies that all the proposed samples of $\boldsymbol{\Theta}$ will be accepted if they satisfy the accept-reject condition $\rho(\mathbf{D}, \mathbf{D}_{obs}) \le \epsilon$.

The "burn-in" and "thinning" of the Markov chains [44,47] can be equivalently applied to the ABC-MCMC method. To monitor convergence of ABC-MCMC, the approach based on the Gelman-Rubin (G-R) statistic [48] can be employed. In this study, we calculate the G-R statistic $\hat{R}$ using three MCMC runs for each parameter, and samples from all three chains are used to make further statistical inferences. A cutoff criterion of $\hat{R} \le 1.01$ is used to determine the iteration length. The reader is referred to [48,49] for more detailed discussions on the G-R statistic.

A possible demerit of the ABC-MCMC scheme is that due to a low acceptance probability (for a very small threshold value) the Markov chain can get trapped initially in low probability regions for a long time, which results in longer burn-in. This problem of burn-in time is addressed in Section 3.1.1.

3.1.1. Initialization of ABC-MCMC

To reduce the burn-in time in ABC-MCMC, we use an efficient initialization scheme (Algorithm 4) proposed by Hazra et al. [50]. The algorithm starts with the selection of a series of tolerance thresholds $\epsilon_1 > \epsilon_2 > \cdots > \epsilon_T$ (ϵ_T is the target tolerance threshold), and sequentially generates the corresponding set of samples $\boldsymbol{\Theta}^{(1)}, \boldsymbol{\Theta}^{(2)}, \cdots, \boldsymbol{\Theta}^{(T)}$. The initial samples $\boldsymbol{\Theta}^{(1)}, \cdots, \boldsymbol{\Theta}^{(T-1)}$ are discarded, and the ABC-MCMC algorithm starts with $\boldsymbol{\Theta}^{(T)}$ as its initial sample. A suitable COV for the initialization proposal distribution $q'(\bullet \to \bullet)$ should be chosen so that the algorithm does not get stuck, i.e., too narrow or too wide support for the proposal distribution should be avoided. For numerical examples, the COV of $q'(\bullet \to \bullet)$ is considered twice the COV of the proposal distribution used in ABC-MCMC algorithm.

1: Initialize $\boldsymbol{\Theta}^{(k)} \sim f(\boldsymbol{\Theta})$; set $k = 1$
2: **for** $k = 1$ to $(n-1)$ **do**
3: Generate $\boldsymbol{\Theta}^*$ from the proposal distribution $q(\boldsymbol{\Theta}^{(k)} \to \boldsymbol{\Theta}^*)$
4: Simulate a data set $\mathbf{D}^*$ from the forward model $\mathcal{M}(\mathbf{D} \mid \boldsymbol{\Theta}^*)$
5: Calculate the distance function $\rho(\mathbf{D}^*, \mathbf{D}_{obs})$
6: Proceed to step 7 if $\rho(\mathbf{D}^*, \mathbf{D}_{obs}) \le \epsilon$; otherwise set $\boldsymbol{\Theta}^{(k+1)} = \boldsymbol{\Theta}^{(k)}$ and go to step 10
7: Calculate the acceptance probability

$$\mathcal{A}(\boldsymbol{\Theta}^{(k)} \to \boldsymbol{\Theta}^*) = \min\left\{1, \frac{f(\boldsymbol{\Theta}^*)q(\boldsymbol{\Theta}^* \to \boldsymbol{\Theta}^{(k)})}{f(\boldsymbol{\Theta}^{(k)})q(\boldsymbol{\Theta}^{(k)} \to \boldsymbol{\Theta}^*)}\right\}$$

8: Simulate u from the uniform distribution $\mathcal{U}[0, 1]$
9: Set

$$\boldsymbol{\Theta}^{(k+1)} = \begin{cases} \boldsymbol{\Theta}^*, & \text{if } u \le \mathcal{A}(\boldsymbol{\Theta}^{(k)} \to \boldsymbol{\Theta}^*) \\ \boldsymbol{\Theta}^{(k)}, & \text{otherwise} \end{cases}$$

10: **end for**

Algorithm 3. ABC-MCMC sampler.

3.1.2. Selection of the tolerance threshold

For the choice of the tolerance threshold, a similar method suggested by Beaumont et al. [23] is adopted in this study. However, the authors proposed the method in an ABC rejection sampler setting, whereas here we apply the same idea in the ABC-MCMC environment. Accordingly, ϵ is set to be a percentile Q_ϵ of the distribution of the proposed distance function values obtained from the proposed moves (samples) of the corresponding MCMC runs. However, the distribution of $\rho(\mathbf{D}, \mathbf{D}_{obs})$ is generally not available a priori; hence, the algorithm can be initialized with a chosen ϵ and later a check can be performed by evaluating whether the chosen ϵ follows the criterion $\mathbb{P}[\rho(\mathbf{D}, \mathbf{D}_{obs}) \le \epsilon] \le Q_\epsilon/100$, where $\mathbb{P}[\bullet]$ represents probability. In this study, a value of $Q_\epsilon = 0.01$ is selected for the choice of tolerance threshold, which means that the chosen ϵ value should be less than the 0.01th percentile of the distribution of $\rho(\mathbf{D}, \mathbf{D}_{obs})$ obtained from the proposed moves by the proposal distribution.

3.1.3. Distance function

To implement the likelihood-free method, we need to select an effective distance function based on the observed degradation data. We propose the following distance function to be used for the numerical study:

$$\rho(\mathbf{D}^*, \mathbf{D}_{obs}) = \|(d_1, d_2, \cdots, d_N)\| \tag{10}$$

where $d_i = \|\Delta\mathbf{y}_i^* - \Delta\mathbf{y}_i\|$. Here, $\|\bullet\|$ is the Euclidean norm (or L^2 norm) operator. This distance function is chosen because the Euclidean norm is a suitable distance measure for data vectors, and minimizing the function given in Eq. (10) helps ABC-MCMC to accept parameter sets closer to the high probability region of the posterior. A flowchart describing the steps of the proposed scheme is presented in Fig. 2.

4. Numerical examples

Four examples are presented to illustrate the application of the proposed likelihood-free approach. The first illustrates the convergence behavior of the ABC posterior, and the second demonstrates the effectiveness of the method in capturing the initial degradation and the non-stationary trend of the degradation process. The third example shows the applicability of the method in estimating the lifetime distribution of a component population. The fourth example is based on practical degradation data collected from a Canadian nuclear reactor. In all four examples, the Bayesian inference schemes are implemented in MATLAB®2017b on a desktop computer with Intel i5-6500 processor.

4.1. Example 1: A one-parameter stationary model

4.1.1. Synthetic data

Assuming zero initial degradation, a synthetic data set consisting of measurements (mm) related to degradation of five fictitious components from three repeated inspections, taken at 5, 10 and 15 years, is simulated from a stationary GP with parameters $\alpha = 4$, $\eta = 1$, and $\beta = 0.015$. The normally distributed sizing error has zero mean and SD $\sigma_Z = 0.1$ mm. Fig. 3 shows the degradation paths of the five components. A similar example can be found in [6].

Select a series of tolerance thresholds $\epsilon_1 > \epsilon_2 > \cdots > \epsilon_T$
Generate $\boldsymbol{\Theta}^{(1)} \sim f_{\epsilon_1}(\boldsymbol{\Theta} \mid \mathbf{D}_{obs})$ using ABC rejection sampler
for $k = 1$ to $(T-1)$ **do**
  repeat
    Generate $\boldsymbol{\Theta}^* \sim q'(\boldsymbol{\Theta}^{(k)} \to \boldsymbol{\Theta}^*)$, such that $f(\boldsymbol{\Theta}^*) \ne 0$
    Simulate $\mathbf{D}^* \sim \mathcal{M}(\mathbf{D} \mid \boldsymbol{\Theta}^*)$
    Accept $\boldsymbol{\Theta}^*$, if $\rho(\mathbf{D}^*, \mathbf{D}_{obs}) \le \epsilon_{k+1}$
  until Acceptance
  Set $\boldsymbol{\Theta}^{(k+1)} = \boldsymbol{\Theta}^*$
end for

Algorithm 4. Initialization of ABC-MCMC.

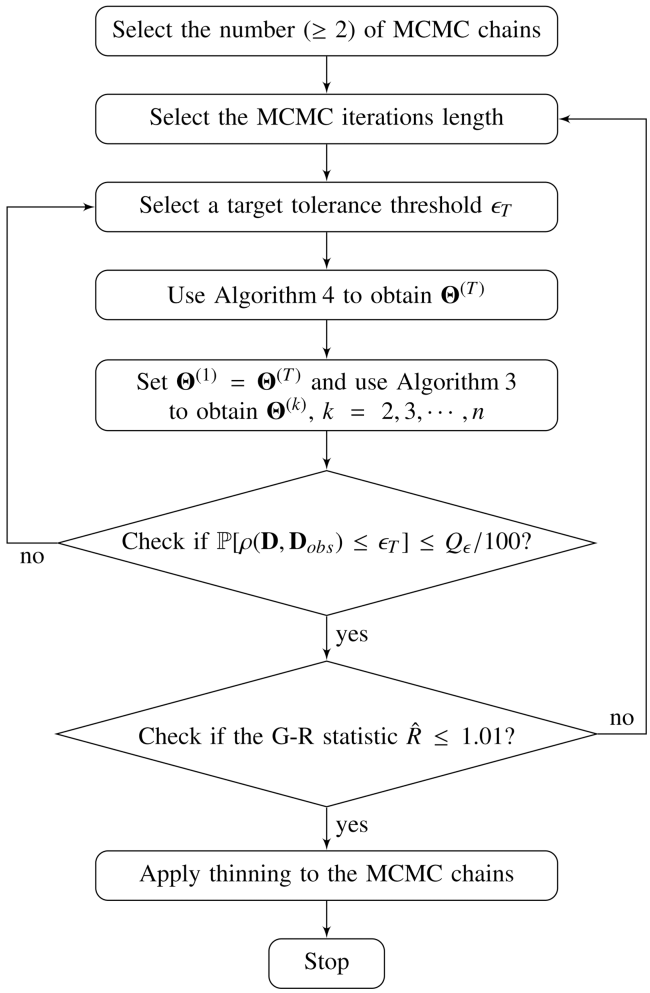

Fig. 2. Flowchart of the proposed ABC-MCMC algorithm.

4.1.2. Numerical analysis

Consider that only parameter α is unknown, and hence, it needs to be estimated. Both ABC-MCMC and L-MCMC methods are implemented for the estimation task. The prior distribution for α is selected to be a uniform distribution, i.e., $f(\alpha) = \mathcal{U}[0, 10]$. The proposal distribution is chosen to be a normal distribution with mean equal to the current sample and COV equal to 0.1 for both methods. In the L-MCMC method, the likelihood function is numerically integrated using Monte Carlo simulation [42] with 500 samples.

For implementing the ABC-MCMC scheme, four values of the tolerance threshold are considered, i.e., $\epsilon = \{0.6, 0.5, 0.4, 0.27\}$. These gradually decreasing tolerance thresholds help to observe the convergence of the ABC posterior. For initializing the ABC-MCMC run, Algorithm 4 is employed. The target threshold ϵ is reached through the following sequence of threshold values: $2\epsilon > 1.75\epsilon > 1.5\epsilon > 1.25\epsilon > \epsilon$. Selection of a thinning interval is performed by plotting sample autocorrelation function (ACF) plots of the marginal Markov chains [44,51].

ABC-MCMC generated three chains for each case as required by the criterion of G-R statistic. For the purpose of representation, Fig. D.14 in Appendix D shows the three MCMC runs and the corresponding sample ACF and G-R statistic plots for the $\epsilon = 0.27$ case. Accordingly, the chosen chain lengths and thinning intervals along with the computation times are presented in Table 1; the computation times are given for single MCMC runs. The sample ACF plots of $\epsilon = 0.27$ case (Fig. D.14) show very high correlation among the samples, and that is why a larger thinning interval of 30,000 is chosen. Although the required length of the MCMC run given by the G-R statistic is only around 2.5×10^6, given the high amount of thinning, a length of 10^7 is selected. The corresponding 0.01th percentile values of the distributions of the proposed distance function values (calculated using proposed samples from all three runs of the algorithm) are 0.282 ($\epsilon = 0.6$), 0.278 ($\epsilon = 0.5$), 0.276 ($\epsilon = 0.4$) and 0.274 ($\epsilon = 0.27$), respectively. It can be noticed that the first three tolerance thresholds, $\epsilon = \{0.6, 0.5, 0.4\}$, are far above the threshold choice criterion $Q_\epsilon \leq 0.01$, whereas, $\epsilon = 0.27 < 0.274$ meets the criterion proposed for tolerance selection (see Section 3.1.2). Fig. 4 shows the distributions of the proposed distance values and the regions of the corresponding chosen tolerance thresholds.

Similarly, L-MCMC chain attributes and the computation time are presented in Table 1. Fig. D.15 in Appendix D shows three MCMC runs and the corresponding plots for sample ACFs and G-R statistic. As can be seen, the sample autocorrelation of the L-MCMC output is much less compared to the likelihood-free approach. The G-R statistic indicates a convergence at around 750 iteration length after burn-in; hence, an MCMC run of 1000 iterations is selected, and the initial 100 samples are discarded to allow burn-in.

Fig. 5 shows the posterior distributions of α produced by both methods using samples from three MCMC runs after burn-in and

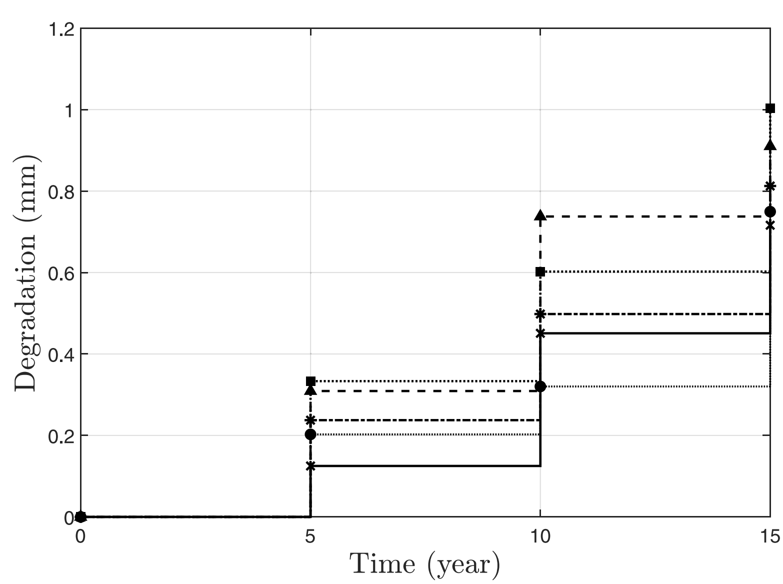

Fig. 3. Observed degradation growth curves of five fictitious components simulated from a stationary GP.

Table 1
Selected attributes of the MCMC chains and computation times.

Method	Tolerance (ϵ)	Chain length	Burn-in length	Thinning interval	Computation time
ABC-MCMC	0.6	1×10^5	–	100	3.3 s
	0.5	2×10^5	–	200	6.4 s
	0.4	2×10^5	–	500	6.4 s
	0.27	1×10^7	–	30,000	320 s
L-MCMC	–	1×10^3	100	3	860 s

thinning. The convergence of the ABC posterior is clearly visible as the tolerance threshold gradually decreases. The ABC posterior with $\epsilon = 0.27$ shows a very good match to the posterior generated using the likelihood-based method. The mean and COV of the ABC posterior with $\epsilon = 0.27$ are 3.65 and 0.096, respectively; whereas the posterior estimated by the likelihood-based method has a mean of 3.66 and COV of 0.078.

This example shows that ABC-MCMC is computationally more efficient in comparison to the L-MCMC method. In fact, we were able to get around 170% more efficiency in the proposed approach than the traditional likelihood-based approach.

4.2. Example 2: A two-parameter non-stationary model

4.2.1. Synthetic data

Degradation paths of ten fictitious components are simulated, and a synthetic measurement data set is generated based on the following parameters: $\alpha = 2$, $\eta = 2.5$, $\beta = 0.01$, $\mu_A = 0.5$ mm, $\sigma_A = 0.1$ mm, and $\sigma_Z = 0.1$ mm. The observations are simulated for three repeat measurements at 2nd, 4th and 6th years of operation; the corresponding degradation paths are shown in Fig. 6.

4.2.2. Numerical analysis

Consider the parameters η and μ_A are unknown, and the goal is to compute the joint and marginal posteriors of these two parameters using the synthetic data set. The prior distributions are considered diffused uniform priors, i.e., $f(\eta, \mu_A) = f(\eta)f(\mu_A)$, where $f(\eta) = \mathcal{U}[0, 5]$ and $f(\mu_A) = \mathcal{U}[0, 1]$. For η and μ_A, the proposal distribution can be conveniently selected as independent normal distributions centered at the current samples with COVs equal to 0.01 and 0.1, respectively.

The ABC-MCMC is employed using $\epsilon = 0.61$ as the tolerance threshold. The initialization of the ABC-MCMC run using Algorithm 4 is conducted through the same sequence of threshold values as used in Simulation example 1. For the L-MCMC method, 5000 Monte Carlo samples are considered for likelihood evaluation.

Both Bayesian inference schemes generated three chains; the chosen chain lengths, burn-in lengths and thinning intervals along with the computation times are presented in Table 2. The computation times are given for single MCMC runs. These chain attributes are selected based on the MCMC traceplots, and their corresponding ACF and G-R statistic plots as shown in Figs. D.16 and D.17 in Appendix D. The 0.01th percentile value estimated from the distribution of the proposed distance values is 0.622, which proves that the chosen threshold $\epsilon = 0.61 < 0.622$ satisfies the tolerance selection criterion.

Fig. 7 depicts the marginal posteriors and scatter plot of the joint posterior of η and μ_A obtained from both Bayesian computation schemes. The means, COVs and 95% credible intervals (CIs) of the marginal posterior distributions are presented in Table 3. The results produced by the likelihood-free method show great similarity to the results obtained from the likelihood-based method. But ABC-MCMC turns out to be extremely efficient in this problem with an overall 400% more efficiency than the likelihood-based method.

4.3. Example 3: The six-parameter non-stationary model

This example is an attempt to estimate all six parameters of the proposed GP model of degradation. The same synthetic data set (used in Example 2) is considered once again. The chosen independent prior distributions are: $f(\alpha) = \mathcal{U}[0, 5]$, $f(\eta) = \mathcal{U}[0, 5]$, $f(\beta) = \mathcal{U}[0, 1]$,

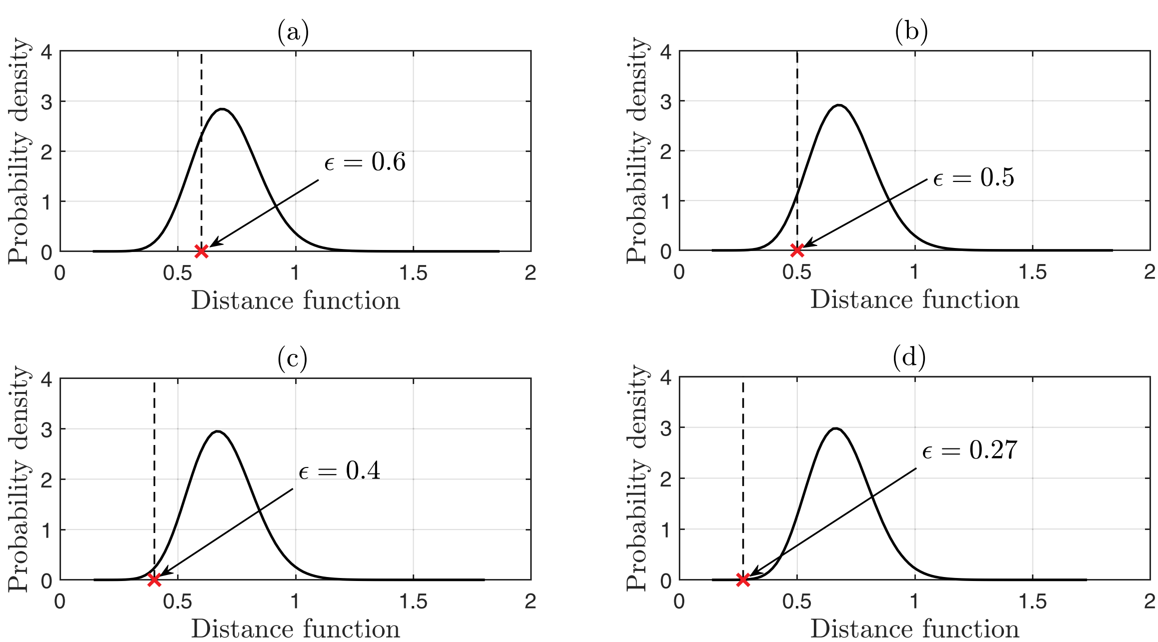

Fig. 4. Distributions of the proposed distance values for (a) $\epsilon = 0.6$, (b) $\epsilon = 0.5$, (c) $\epsilon = 0.4$, and (d) $\epsilon = 0.27$ cases. Tolerance threshold $\epsilon = 0.27$ satisfies the tolerance selection criterion.

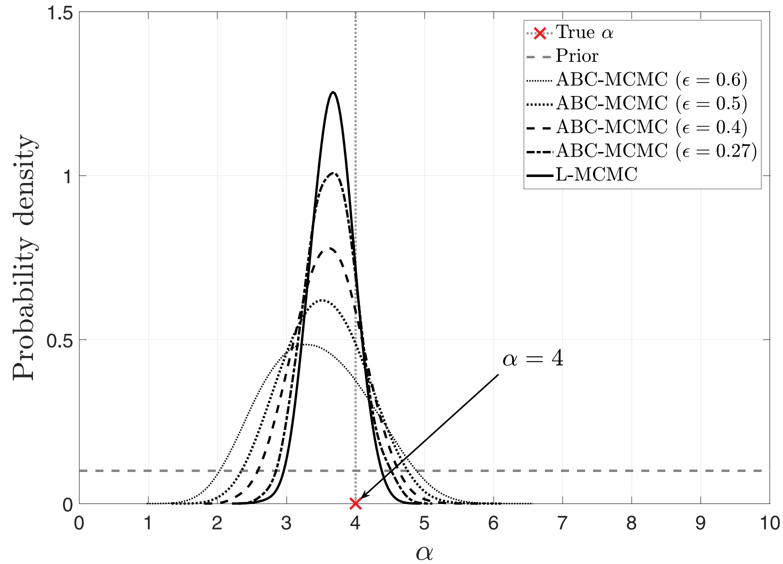

Fig. 5. Posterior distributions of α produced by ABC-MCMC and L-MCMC.

$f(\mu_A) = \mathcal{U}[0, 1]$, $f(\sigma_A) = \mathcal{U}[0, 1]$, and $f(\sigma_Z) = \mathcal{U}[0, 1]$. Independent normal distributions with means equal to current samples and COVs equal to 0.1 are chosen as proposal distributions. For ABC-MCMC, the initial proposal distributions are chosen to have twice the COVs of the proposals used in the main algorithm. A tolerance threshold of $\epsilon = 0.52$ is selected for ABC-MCMC. Due to a high dimension of the parameter space, the acceptance rate of both methods is found to be very low, thus resulting in very long MCMC chains. To remedy this, the model parameters are updated one-by-one sequentially; this reduces the rejection rate in both methods.

The likelihood function in L-MCMC is integrated using 5000 Monte Carlo samples. L-MCMC spent around 7 hours to generate a single chain of 3000 iterations length. A total of three MCMC chains are generated, and all of them converged at around 1700 iterations; hence a burn-in length of the same amount was chosen. Similarly, ABC-MCMC generated a single chain of 5×10^7 iterations length in around 2 hours and 40 minutes.

The marginal posterior distributions of the model parameters are shown in Fig. 8 and their summary statistics are presented in Table 4. It is seen that ABC-MCMC and L-MCMC produced comparable results. Marginal posteriors of α produced by both methods seems to be very close; however, a noticeable difference in posterior locations can be spotted for parameters η, β, and μ_A. The reason behind this anomaly is the selected threshold value not being zero, resulting in an approximation of the joint posterior. Moreover, it can be noticed that the standard deviation parameters, σ_A and σ_Z, are difficult to estimate using both the methods.

To estimate the lifetime distribution of the component population, a critical limit of 4 mm degradation is selected. With the help of simulation, the mean and fifth percentile of population lifetime are estimated and are presented in Fig. 9. The statistical properties of the lifetime distribution are tabulated in Table 5. The true values of the lifetime quantiles are estimated using the true model parameter values. Both the likelihood-based and likelihood-free approaches produced almost similar distributions for the mean and fifth percentile of lifetime estimates. Moreover, both methods show excellent accuracy in comparison to the estimates obtained using the true values of the model parameters.

Overall, given only a small data set, it is found that both methods perform reasonably well in estimating all the six parameters, and

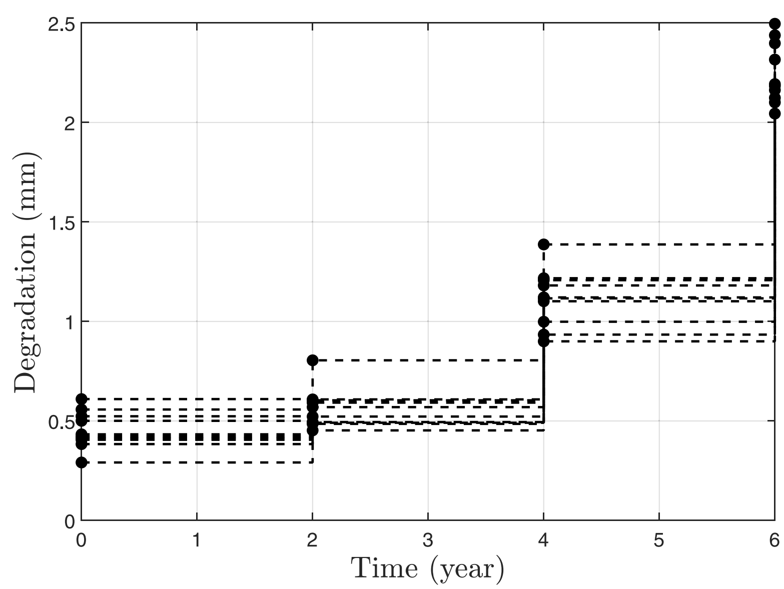

Fig. 6. Observed degradation growth curves of ten fictitious components simulated from a non-stationary GP.

Table 2
Selected attributes of the MCMC chains and computation times.

Method	Tolerance (ϵ)	Chain length	Burn-in length	Thinning interval	Computation time
ABC-MCMC	0.61	2×10^7	–	50,000	1790 s
L-MCMC	–	1×10^3	100	4	9125 s

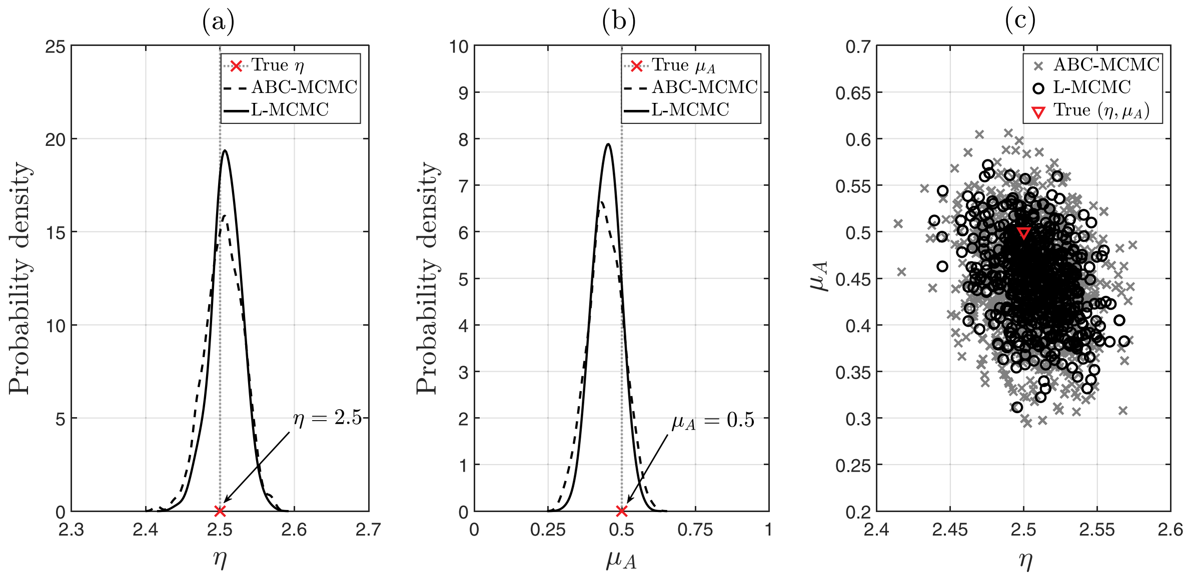

Fig. 7. Marginal posterior distributions of (a) η and (b) μ_A and (c) the joint posterior distributions as two-dimensional scatter plots produced by ABC-MCMC and L-MCMC.

provide highly accurate estimates of the lifetime quantiles. Once again, ABC-MCMC proves to be highly efficient with 160% more efficiency than L-MCMC.

4.4. Practical case study

The piping network of the primary heat transport system of Canada Deuterium Uranium (CANDU®) reactors is subject to degradation due to their operation under high temperature and pressure. The outlet feeder pipes mostly suffer from wall thickness loss due to erosion by the flow-accelerated corrosion (FAC) mechanism [5,45,52]. To assess the extent of degradation and schedule necessary pipe replacements, different pipe sections are repeatedly inspected at various time intervals. The inspection data are further utilized in degradation model calibration, degradation prediction, and scheduling of maintenance operations. The proposed likelihood-free Bayesian inference approach is implemented to solve the parameter estimation problem for the GP model using practical data of wall thinning of feeder pipes from a CANDU®nuclear reactor.

The inspection data related to minimum wall thicknesses of 37 feeder pipes are considered in this study. These pipes have a common diameter of 2” and nominal thickness of 5.5 mm. Fig. 10 shows the degradation paths of the feeder pipes, i.e., each dotted line represents the wall thickness losses over time of a different pipe derived from the minimum wall thickness measurements. These feeder pipes were inspected two to five times using an ultrasonic probe between 8.55 and 22.25 effective full power years (EFPY).

To estimate the GP model parameters using ABC-MCMC, the following prior distributions are chosen: $f(\alpha) = \mathcal{U}[0, 50]$, $f(\eta) = \mathcal{U}[0, 5]$, $f(\beta) = \mathcal{U}[0, 1]$, $f(\mu_A) = \mathcal{U}[0, 5]$, $f(\sigma_A) = \mathcal{U}[0, 1]$, and $f(\sigma_Z) = \mathcal{U}[0, 1]$. The marginal posterior distributions of the model parameters are shown in Fig. 11. The means, COVs, and 95% CIs of the parameters are presented in Table 6. The mean of the time parameter η is found to be around 1.6, which proves that the underlying degradation process is mostly non-stationary. The COV values reflect high uncertainties in the model parameters. However, the mean initial degradation parameter μ_A and the time parameter η show less uncertainties compared to other model parameters.

Fig. 12 shows the predicted mean degradation path along with the 95% CI. The distributions of mean and fifth percentile of the population lifetime are presented in Fig. 13 and their statistical properties are presented in Table 7. Accordingly, if a wall thickness loss of 60% of the nominal thickness is considered to be a critical limit, then we can expect the feeder pipes to reach their end of life by around 37.39 EFPY. The lifetime distributions show that both mean and fifth percentile lifetimes possess high uncertainty, which could result from limited

Table 3
Summary statistics of the posterior parameter distributions.

Parameter	True value	ABC-MCMC			L-MCMC		
		Mean	COV	95% CI	Mean	COV	95% CI
η	2.5	2.5052	0.0101	[2.4540, 2.5513]	2.5085	0.0081	[2.4652, 2.5472]
μ_A	0.5	0.4448	0.1312	[0.3296, 0.5559]	0.4485	0.0995	[0.3625, 0.5332]

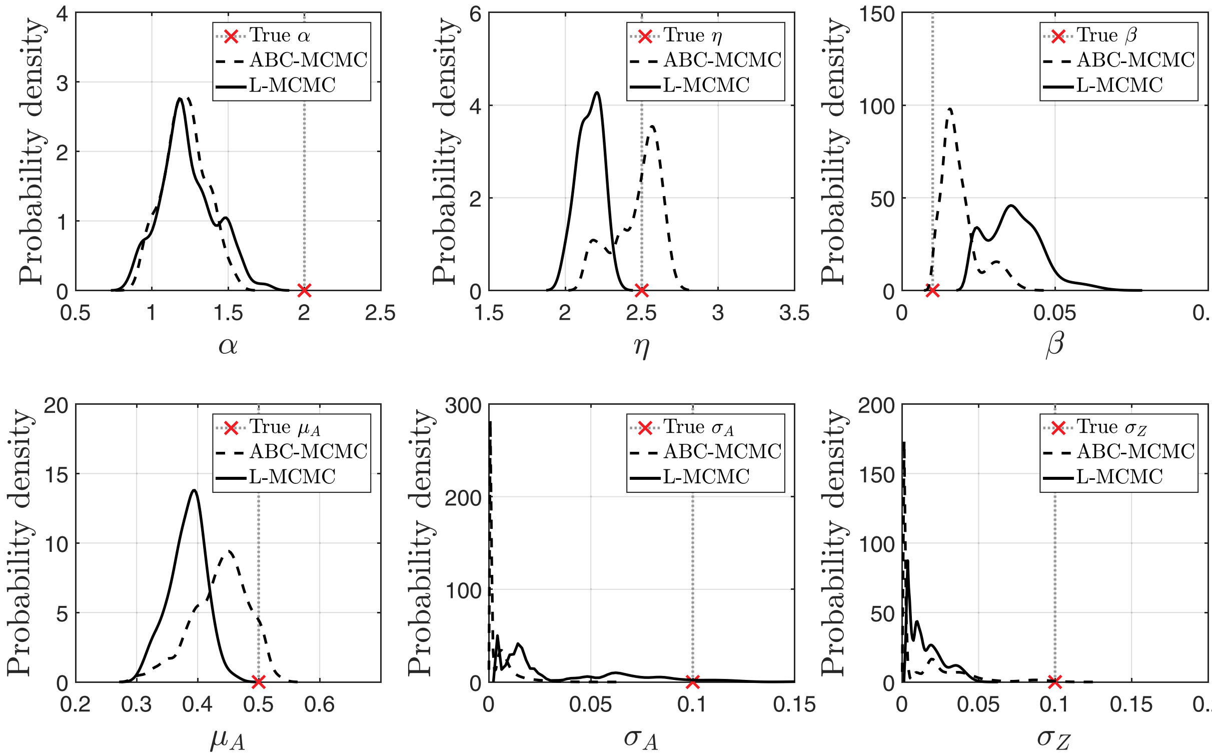

Fig. 8. Marginal posterior distributions of the model parameters produced by ABC-MCMC and L-MCMC.

Table 4
Summary statistics of the posterior parameter distributions.

Parameter	True value	ABC-MCMC			L-MCMC		
		Mean	COV	95% CI	Mean	COV	95% CI
α	2	1.2171	0.1144	[0.9583, 1.4901]	1.2352	0.1462	[0.9172, 1.5842]
η	2.5	2.4693	0.0614	[2.1595, 2.6560]	2.1603	0.0357	[2.0046, 2.2922]
β	0.01	0.0187	0.3217	[0.0109, 0.0335]	0.0364	0.2401	[0.0227, 0.0566]
μ_A	0.5	0.4366	0.1056	[0.3268, 0.5118]	0.3831	0.0780	[0.3218, 0.4370]
σ_A	0.1	0.0032	1.6477	[0.0010, 0.0200]	0.0376	0.9049	[0.0035, 0.1168]
σ_Z	0.1	0.0139	1.4748	[0.0012, 0.0880]	0.0160	0.6996	[0.0025, 0.0403]

observations. A quantity of interest generally used in maintenance decision making is the fifth percentile of fifth percentile lifetime, $T_{05/05}$, which is found to be around 28.74 EFPY.

5. Conclusions

In this paper, the Bayesian inference of the GP parameters from noisy degradation measurements is performed through a novel application of the ABC-MCMC method. Large sampling and inspection uncertainties in degradation measurements require uncertainty quantification of the GP model parameters in a Bayesian setting. However, traditional likelihood-based Bayesian inference schemes are computationally intensive because the sample likelihood of the model parameters derived from noisy data is a complicated high-dimensional integration. The proposed ABC-MCMC method can be implemented with confidence to completely bypass the likelihood evaluation step in the Bayesian inference procedure. From this study, the following conclusions can be derived:

1. ABC-MCMC is a special case of the MH algorithm, where the target distribution of the sampler is the ABC posterior instead of the true posterior.
2. The ABC posterior converges to the true posterior as the tolerance of the analysis tightened.
3. The tolerance selection criterion proposed in this study provides a general framework to perform a quality check on the accepted samples generated by the ABC-MCMC scheme, which guarantees that the accepted samples belong to a distribution very close to the true posterior distribution. The proposed criterion is validated through various numerical examples by comparing the accuracy of ABC-MCMC with the traditional likelihood-based approach.
4. The samples from the Markov chain of the proposed method show high correlation; to reduce the sample correlation, a large thinning interval should be chosen.
5. The ABC-MCMC method combined with the proposed initialization scheme is highly efficient compared to the traditional likelihood-based method. However, the computational efficiency depends solely upon the type and complexity of the problem. For instance, in the one-parameter estimation problem, the proposed method shows 170% more efficiency than the traditional method, whereas in the two-parameter estimation problem, it shows an efficiency as high as 400% more than the likelihood-based method.

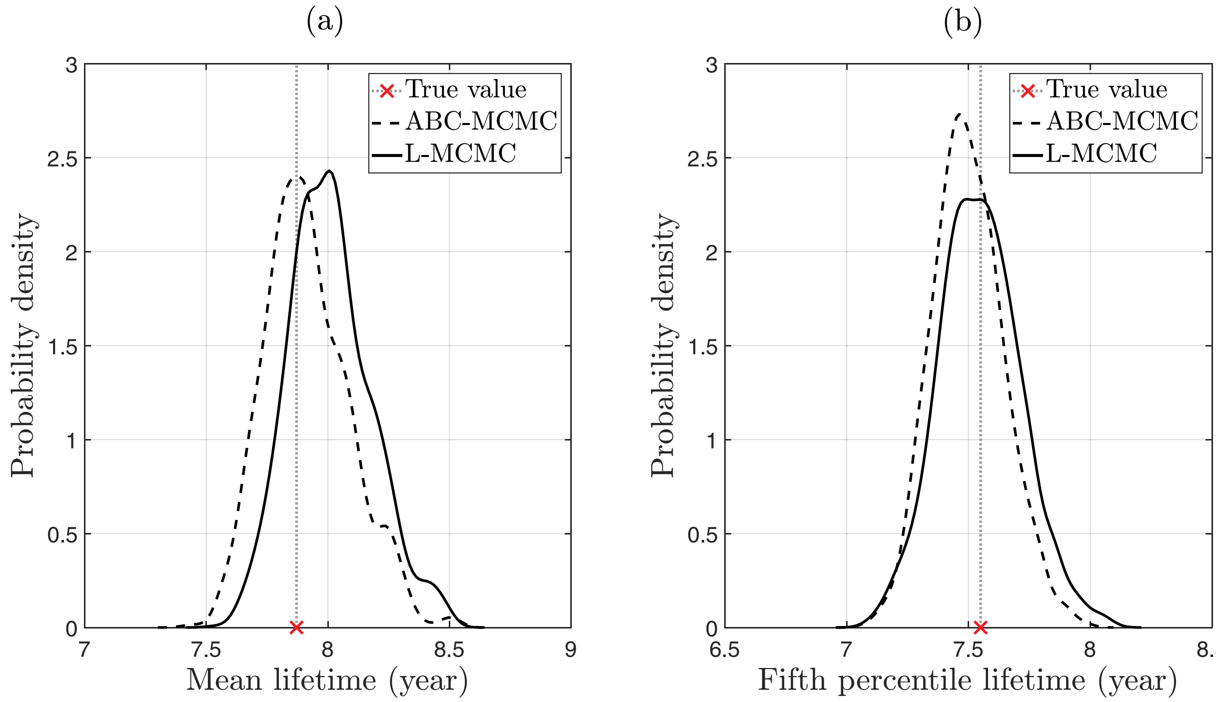

Fig. 9. (a) Mean and (b) fifth percentile of the lifetime distribution produced by ABC-MCMC and L-MCMC.

Table 5
Statistical properties of the lifetime distribution.

Lifetime	True value (year)	ABC-MCMC Mean (year)	ABC-MCMC COV	ABC-MCMC 95% CI	L-MCMC Mean (year)	L-MCMC COV	L-MCMC 95% CI
Mean	7.87	7.91	0.0219	[7.62, 8.28]	8.01	0.0208	[7.71, 8.40]
Fifth percentile	7.55	7.49	0.0195	[7.25, 7.80]	7.55	0.0221	[7.23, 7.89]

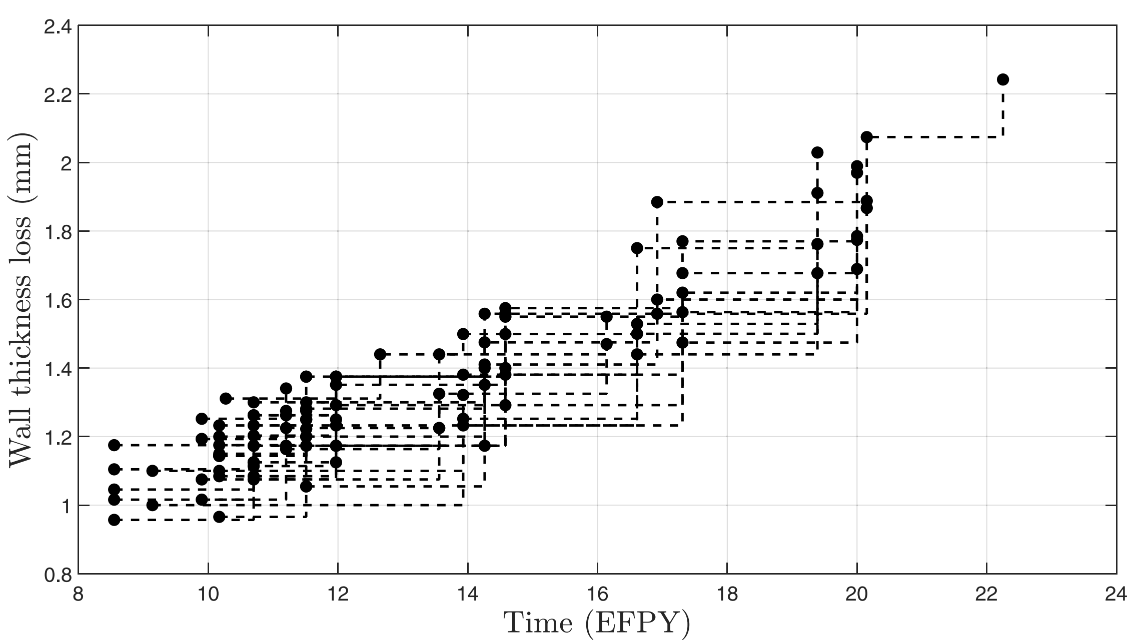

Fig. 10. Observed wall thickness losses of the 37 inspected feeder pipes.

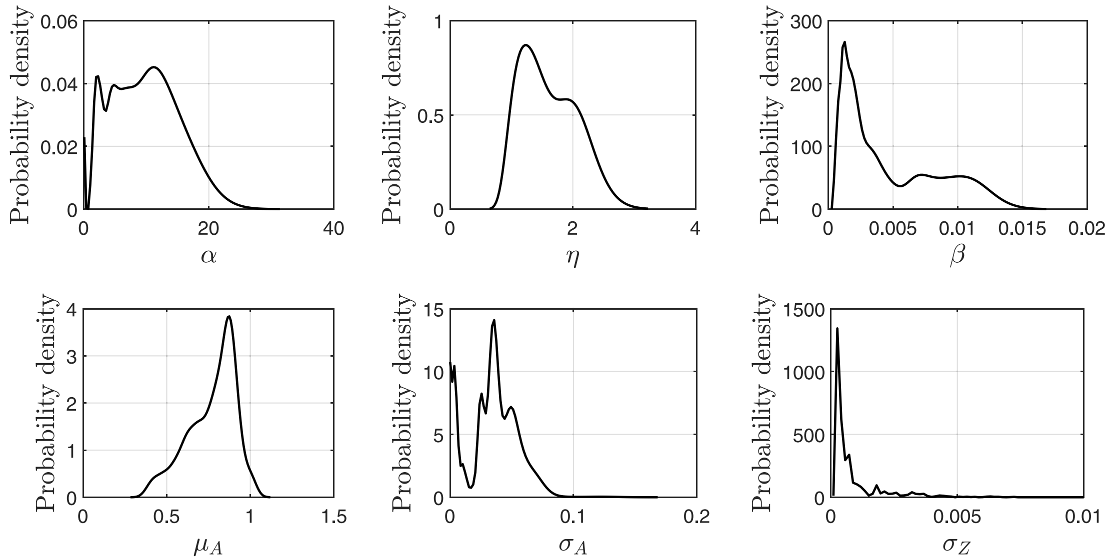

Fig. 11. Marginal posterior distributions of the GP model parameters.

Table 6
Statistical properties of the estimated parameters.

Parameter	Mean	COV	95% CI
α	6.7161	0.9155	[0.1358, 19.012]
η	1.5985	0.2504	[1.0122, 2.1135]
β	0.0048	0.7678	[0.0007, 0.0118]
μ_A	0.7715	0.1841	[0.4375, 0.9902]
σ_A	0.0153	1.4075	[0.0001, 0.0642]
σ_Z	0.0007	1.7409	[0.0001, 0.0037]

6. When the number of parameters to be estimated increases, ABC-MCMC may result in slightly erroneous estimates for the marginal posteriors, nevertheless, it still produces highly accurate estimates of the lifetime quantiles.
7. The case study involving corrosion modelling and estimation of population lifetime of a nuclear piping system shows that the proposed method can be implemented to circumvent the complications which arise in likelihood formulation and its evaluation in practical scenarios.

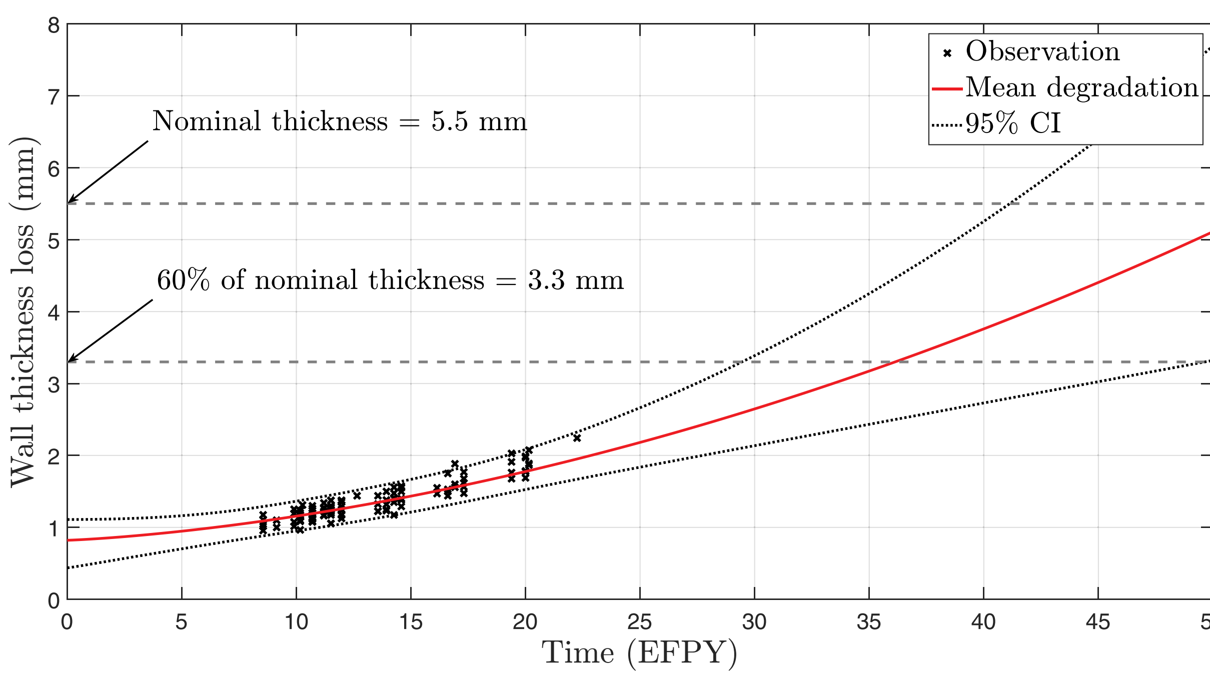

Fig. 12. Predicted mean degradation growth with 95% credible interval.

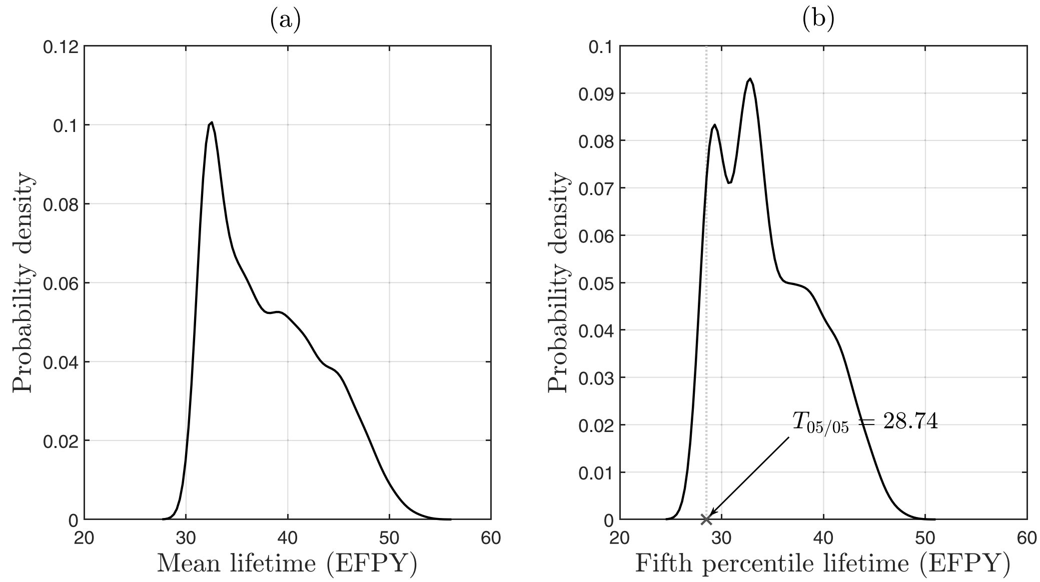

Fig. 13. (a) Mean and (b) fifth percentile of the lifetime distribution.

Table 7
Statistical properties of the lifetime distribution.

Lifetime	Mean (EFPY)	COV	95% CI
Mean	37.39	0.1381	[31.36, 48.73]
Fifth percentile	33.57	0.1351	[27.61, 44.15]

Author statement

We, all authors, hereby confirm that we don't have any financial and personal relationships with other people or organizations that could inappropriately influence (bias) our work.

We have nothing to disclose.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Likelihood function for noisy data

Suppose degradation monitoring data are collected from repeated inspections from a group of N components at various time intervals. A basic assumption in stochastic degradation modelling is that the GP parameters, α, η and β, are constant for all the components subjected to degradation. For an ith component, true values (noise-free) of degradation at different inspection times $t_{i0}, t_{i1}, \cdots, t_{im_i}$ are denoted as $x_{i0}, x_{i1}, \cdots, x_{im_i}$, respectively, where $x_{i0} = 0$ at time $t_{i0} = 0$, and m_i is the total number of inspections. The degradation growth over a time interval $\Delta t_{ij} = t_{ij} - t_{i,j-1}$ is denoted as $\Delta x_{ij} = x_{ij} - x_{i,j-1}$, where $j = 1, 2, \cdots, m_i$. Here, Δx_{ij} can be assumed to be a realization of the random variable ΔX_{ij} which follows the gamma distribution $\mathcal{G}(\bullet|\alpha\Delta t_{ij}^{(\eta)}, \beta)$, where $\Delta t_{ij}^{(\eta)} = (t_{ij}^{\eta} - t_{i,j-1}^{\eta})$. All degradation increments for an ith component are denoted as a vector, $\Delta\mathbf{x}_i = \{\Delta x_{i1}, \Delta x_{i2}, \cdots, \Delta x_{im_i}\}^T$, with sampling distribution $f_{\Delta\mathbf{X}_i}(\Delta\mathbf{x}_i) = \prod_{j=1}^{m_i} \mathcal{G}(\Delta x_{ij}|\alpha\Delta t_{ij}^{(\eta)},\beta)$.

The sample likelihood for degradation measurement data (assuming y_{ij} is a realization of the random variable Y_{ij}) collected from an ith component can be written in terms of the joint distribution of $Y_{i1}, Y_{i2}, \cdots, Y_{im_i}$, which are dependent variables. Following Lu [1] and Yuan [2], it is more convenient to write the sample likelihood in terms of joint distribution of the degradation increments. Let a measured degradation increment for an ith component over a time interval Δt_{ij} be denoted as $\Delta y_{ij} = y_{ij} - y_{i,j-1}$, and the corresponding increment in the noise is $\Delta z_{ij} = z_{ij} - z_{i,j-1}$. Measured values of incremental degradation and corresponding increment in noise are denoted in vector forms as $\Delta\mathbf{y}_i = \{\Delta y_{i1}, \Delta y_{i2}, \cdots, \Delta y_{im_i}\}^T$ and $\Delta\mathbf{z}_i = \{\Delta z_{i1}, \Delta z_{i2}, \cdots, \Delta z_{im_i}\}^T$. Note that $\Delta y_{i1} = y_{i1} - y_{i0}$ is unknown because $y_{i0} = a_i$ is a latent variable. The quantities $\Delta\mathbf{y}_i$ and $\Delta\mathbf{z}_i$ are assumed to be realizations of the random vectors $\Delta\mathbf{Y}_i$ and $\Delta\mathbf{Z}_i$, respectively. Now, the joint PDF of $\Delta\mathbf{Y}_i$ can be written through a convolution integral as

$$f_{\Delta\mathbf{Y}_i}(\Delta\mathbf{y}_i) = \int_{a_i}\int_{\Delta\mathbf{z}_i} f_{\Delta\mathbf{X}_i}(\Delta\mathbf{y}_i - \Delta\mathbf{z}_i|a_i) f_A(a_i) f_{\Delta\mathbf{Z}_i}(\Delta\mathbf{z}_i)\, da_i\, d(\Delta\mathbf{z}_i) \tag{A.1}$$

This convolution integral consists of the joint PDFs $f_{\Delta\mathbf{X}_i}(\Delta\mathbf{x}_i)$ and $f_{\Delta\mathbf{Z}_i}(\Delta\mathbf{z}_i)$, where $f_{\Delta\mathbf{Z}_i}(\Delta\mathbf{z}_i)$ is the joint PDF of the random vector $\Delta\mathbf{Z}_i$, which follows the multivariate normal distribution given as (see [6] for the derivation)

$$f_{\Delta\mathbf{Z}_i}(\Delta\mathbf{z}_i) = \frac{1}{(2\pi)^{m_i/2}|\Sigma_{\Delta\mathbf{Z}_i}|}\exp\left\{-\frac{1}{2}\Delta\mathbf{z}_i\Sigma_{\Delta\mathbf{Z}_i}^{-1}\Delta\mathbf{z}_i^T\right\} \tag{A.2}$$

where $|\bullet|$ represents the determinant operator and $\Sigma_{\Delta\mathbf{Z}_i}$ represents the variance-covariance matrix of the random vector $\Delta\mathbf{Z}_i$.

Assuming that the degradation process is independent across the component population, the sample likelihood function for measured degradation data collected from all N components is written as

$$\mathcal{L}(\alpha, \eta, \beta, \mu_A, \sigma_A, \sigma_Z | \Delta \mathbf{y}_1, \Delta \mathbf{y}_2, \cdots, \Delta \mathbf{y}_N) = \prod_{i=1}^{N} f_{\Delta \mathbf{Y}_i}(\Delta \mathbf{y}_i) \tag{A.3}$$

Appendix B. Convergence of ABC posterior

The Bayesian statement for the joint ABC posterior can be written as

$$f(\boldsymbol{\Theta}, \mathbf{D} | \rho(\mathbf{D}, \mathbf{D}_{obs}) \le \epsilon) = \frac{f(\rho(\mathbf{D}, \mathbf{D}_{obs}) \le \epsilon | \boldsymbol{\Theta}, \mathbf{D}) \mathcal{M}(\mathbf{D}|\boldsymbol{\Theta}) f(\boldsymbol{\Theta})}{\int_{\boldsymbol{\Theta}} \int_{\mathbf{D}} f(\rho(\mathbf{D}, \mathbf{D}_{obs}) \le \epsilon | \boldsymbol{\Theta}, \mathbf{D}) \mathcal{M}(\mathbf{D}|\boldsymbol{\Theta}) f(\boldsymbol{\Theta}) d\boldsymbol{\Theta} d\mathbf{D}} \tag{B.1}$$

Now, using Eq. (8), the marginal ABC posterior of $\boldsymbol{\Theta}$ can be derived as

$$\begin{aligned} f_\epsilon(\boldsymbol{\Theta}|\mathbf{D}_{obs}) &= \int_{\mathbf{D}} f(\boldsymbol{\Theta}, \mathbf{D} | \rho(\mathbf{D}, \mathbf{D}_{obs}) \le \epsilon) d\mathbf{D} \\ &= \frac{\left\{ \int_{\mathbf{D}} f(\rho(\mathbf{D}, \mathbf{D}_{obs}) \le \epsilon | \boldsymbol{\Theta}, \mathbf{D}) \mathcal{M}(\mathbf{D}|\boldsymbol{\Theta}) d\mathbf{D} \right\} f(\boldsymbol{\Theta})}{\int_{\boldsymbol{\Theta}} \int_{\mathbf{D}} f(\rho(\mathbf{D}, \mathbf{D}_{obs}) \le \epsilon | \boldsymbol{\Theta}, \mathbf{D}) \mathcal{M}(\mathbf{D}|\boldsymbol{\Theta}) f(\boldsymbol{\Theta}) d\boldsymbol{\Theta} d\mathbf{D}} \\ &= \frac{\mathcal{L}_\epsilon(\boldsymbol{\Theta}|\mathbf{D}_{obs}) f(\boldsymbol{\Theta})}{\int_{\boldsymbol{\Theta}} \mathcal{L}_\epsilon(\boldsymbol{\Theta}|\mathbf{D}_{obs}) f(\boldsymbol{\Theta}) d\boldsymbol{\Theta}} \end{aligned} \tag{B.2}$$

where $\mathcal{L}_\epsilon(\boldsymbol{\Theta}|\mathbf{D}_{obs})$ is the approximate likelihood function used by the ABC algorithm. The approximate likelihood function can be further simplified as

$$\begin{aligned} \mathcal{L}_\epsilon(\boldsymbol{\Theta}|\mathbf{D}_{obs}) &= \int_{\mathbf{D}} f(\rho(\mathbf{D}, \mathbf{D}_{obs}) \le \epsilon | \boldsymbol{\Theta}, \mathbf{D}) \mathcal{M}(\mathbf{D}|\boldsymbol{\Theta}) d\mathbf{D} \\ &= \int_{\mathbf{D}} f(\rho(\mathbf{D}, \mathbf{D}_{obs}) | \boldsymbol{\Theta}, \mathbf{D}) \mathbb{I}_\epsilon[\rho(\mathbf{D}, \mathbf{D}_{obs})] \mathcal{M}(\mathbf{D}|\boldsymbol{\Theta}) d\mathbf{D} \end{aligned} \tag{B.3}$$

where $\mathbb{I}_\epsilon[\,\bullet\,]$ is an indicator function:

$$\mathbb{I}_\epsilon[\rho(\mathbf{D}, \mathbf{D}_{obs})] = \begin{cases} 1, & \text{if } \rho(\mathbf{D}, \mathbf{D}_{obs}) \le \epsilon \\ 0, & \text{otherwise} \end{cases}$$

Note that if $\epsilon \to \infty$, the approximate posterior converges to the prior, i.e.,

$$f_\epsilon(\boldsymbol{\Theta} \mid \mathbf{D}_{obs}) \xrightarrow{\epsilon \to \infty} f(\boldsymbol{\Theta}) \tag{B.4}$$

and if $\epsilon \to 0$, then the ABC posterior will converge to the exact posterior, i.e.,

$$f_\epsilon(\boldsymbol{\Theta} \mid \mathbf{D}_{obs}) \xrightarrow{\epsilon \to 0} f(\boldsymbol{\Theta} \mid \mathbf{D}_{obs}) \tag{B.5}$$

Appendix C. ABC-MCMC is a special case of the MH sampler

The goal of the ABC-MCMC method is to draw samples from the approximate marginal posterior $f_\epsilon(\boldsymbol{\Theta}|\mathbf{D}_{obs})$ by generating a Markov chain that has the same marginal stationary distribution. For the target joint ABC posterior $f(\boldsymbol{\Theta}, \mathbf{D}|\rho(\mathbf{D}, \mathbf{D}_{obs}) \le \epsilon)$, the acceptance probability of an MH sampler can be written as

$$\begin{aligned} &\mathcal{A}(\{\boldsymbol{\Theta}, \mathbf{D}\} \to \{\boldsymbol{\Theta}^*, \mathbf{D}^*\}) \\ &= \min\left\{1, \frac{f(\boldsymbol{\Theta}^*, \mathbf{D}^* | \rho(\mathbf{D}^*, \mathbf{D}_{obs}) \le \epsilon) q_p(\{\boldsymbol{\Theta}^*, \mathbf{D}^*\} \to \{\boldsymbol{\Theta}, \mathbf{D}\})}{f(\boldsymbol{\Theta}, \mathbf{D} | \rho(\mathbf{D}, \mathbf{D}_{obs}) \le \epsilon) q_p(\{\boldsymbol{\Theta}, \mathbf{D}\} \to \{\boldsymbol{\Theta}^*, \mathbf{D}^*\})}\right\} \\ &\text{Substituting Eq. (B.1)} \\ &= \min\left\{1, \frac{f(\rho(\mathbf{D}^*, \mathbf{D}_{obs}) \le \epsilon | \boldsymbol{\Theta}^*, \mathbf{D}^*) \mathcal{M}(\mathbf{D}^*|\boldsymbol{\Theta}^*) f(\boldsymbol{\Theta}^*) q_p(\{\boldsymbol{\Theta}^*, \mathbf{D}^*\} \to \{\boldsymbol{\Theta}, \mathbf{D}\})}{f(\rho(\mathbf{D}, \mathbf{D}_{obs}) \le \epsilon | \boldsymbol{\Theta}, \mathbf{D}) \mathcal{M}(\mathbf{D}|\boldsymbol{\Theta}) f(\boldsymbol{\Theta}) q_p(\{\boldsymbol{\Theta}, \mathbf{D}\} \to \{\boldsymbol{\Theta}^*, \mathbf{D}^*\})}\right\} \\ &= \min\left\{1, \frac{f(\rho(\mathbf{D}^*, \mathbf{D}_{obs}) | \boldsymbol{\Theta}^*, \mathbf{D}^*) \mathbb{I}_\epsilon[\rho(\mathbf{D}^*, \mathbf{D}_{obs})] \mathcal{M}(\mathbf{D}^*|\boldsymbol{\Theta}^*) f(\boldsymbol{\Theta}^*) q_p(\{\boldsymbol{\Theta}^*, \mathbf{D}^*\} \to \{\boldsymbol{\Theta}, \mathbf{D}\})}{f(\rho(\mathbf{D}, \mathbf{D}_{obs}) | \boldsymbol{\Theta}, \mathbf{D}) \mathbb{I}_\epsilon[\rho(\mathbf{D}, \mathbf{D}_{obs})] \mathcal{M}(\mathbf{D}|\boldsymbol{\Theta}) f(\boldsymbol{\Theta}) q_p(\{\boldsymbol{\Theta}, \mathbf{D}\} \to \{\boldsymbol{\Theta}^*, \mathbf{D}^*\})}\right\} \end{aligned} \tag{C.1}$$

where $q_p(\{\bullet, \bullet\} \to \{\bullet, \bullet\})$ is the proposal distribution. Now, ABC-MCMC proposes a move by dividing the proposal function $q_p(\{\boldsymbol{\Theta}, \mathbf{D}\} \to \{\boldsymbol{\Theta}^*, \mathbf{D}^*\})$ into four conditionally independent steps:

1. Propose a move from $\boldsymbol{\Theta}$ to $\boldsymbol{\Theta}^*$ using a proposal distribution $q(\boldsymbol{\Theta} \to \boldsymbol{\Theta}^*)$.
2. Simulate a data set $\mathbf{D}^*$ from the forward model $\mathcal{M}(\mathbf{D}|\boldsymbol{\Theta}^*)$.
3. Calculate the distance function $\rho(\mathbf{D}^*, \mathbf{D}_{obs})$.
4. Proceed if $\rho(\mathbf{D}^*, \mathbf{D}_{obs}) \le \epsilon$; otherwise reject $\boldsymbol{\Theta}^*$, and stay at $\boldsymbol{\Theta}$.

These steps divide the proposal function $q_p(\{\boldsymbol{\Theta}, \mathbf{D}\} \to \{\boldsymbol{\Theta}^*, \mathbf{D}^*\})$ into four conditionally independent components, i.e.,

$$\begin{aligned} &q_p(\{\boldsymbol{\Theta}, \mathbf{D}\} \to \{\boldsymbol{\Theta}^*, \mathbf{D}^*\}) = \\ &\qquad q(\boldsymbol{\Theta} \to \boldsymbol{\Theta}^*) \mathcal{M}(\mathbf{D}^*|\boldsymbol{\Theta}^*) f(\rho(\mathbf{D}^*, \mathbf{D}_{obs}) | \boldsymbol{\Theta}^*, \mathbf{D}^*) \mathbb{I}_\epsilon[\rho(\mathbf{D}, \mathbf{D}_{obs})] \end{aligned} \tag{C.2}$$

Now, if we substitute Eq. (C.2) into Eq. (C.1), we will get the acceptance probability of the ABC-MCMC algorithm:

$$\mathcal{A}(\boldsymbol{\Theta} \to \boldsymbol{\Theta}^*) = \min\left\{1, \frac{f(\boldsymbol{\Theta}^*)q(\boldsymbol{\Theta}^* \to \boldsymbol{\Theta})}{f(\boldsymbol{\Theta})q(\boldsymbol{\Theta} \to \boldsymbol{\Theta}^*)}\right\} \tag{C.3}$$

Note that it is perfectly reasonable to write the acceptance probability as $\mathcal{A}(\boldsymbol{\Theta} \to \boldsymbol{\Theta}^*)$, instead of $\mathcal{A}(\{\boldsymbol{\Theta}, \mathbf{D}\} \to \{\boldsymbol{\Theta}^*, \mathbf{D}^*\})$, because $\mathbf{D}^*$ is simulated from a model conditioned on $\boldsymbol{\Theta}^*$. This derivation proves that ABC-MCMC is a special case of the MH sampler, where the stationary distribution of the Markov chain is the ABC posterior $f_\epsilon(\boldsymbol{\Theta}|\mathbf{D}_{obs})$.

Appendix D. MCMC trace plots and corresponding sample ACF and G-R statistic convergence plots

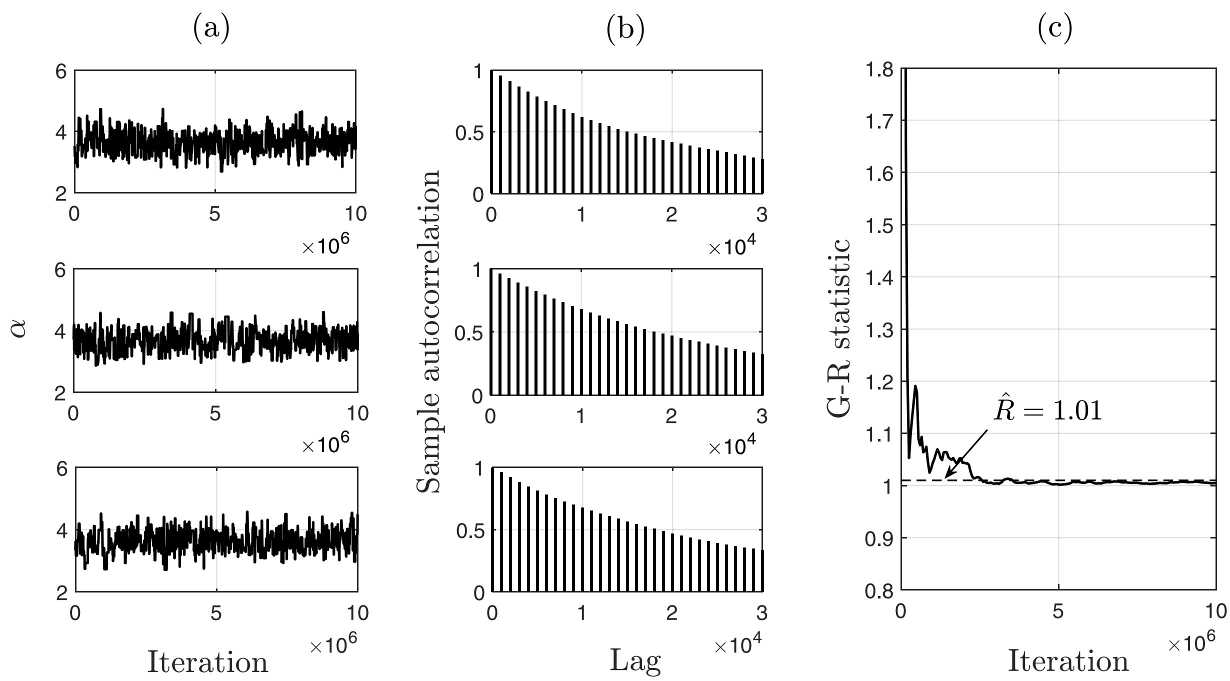

Fig. D1. (a) Three MCMC runs of the parameter α generated by ABC-MCMC with $\epsilon = 0.27$, and the corresponding (b) sample ACF plots, and (c) the convergence of the G-R statistic (Section 4.1).

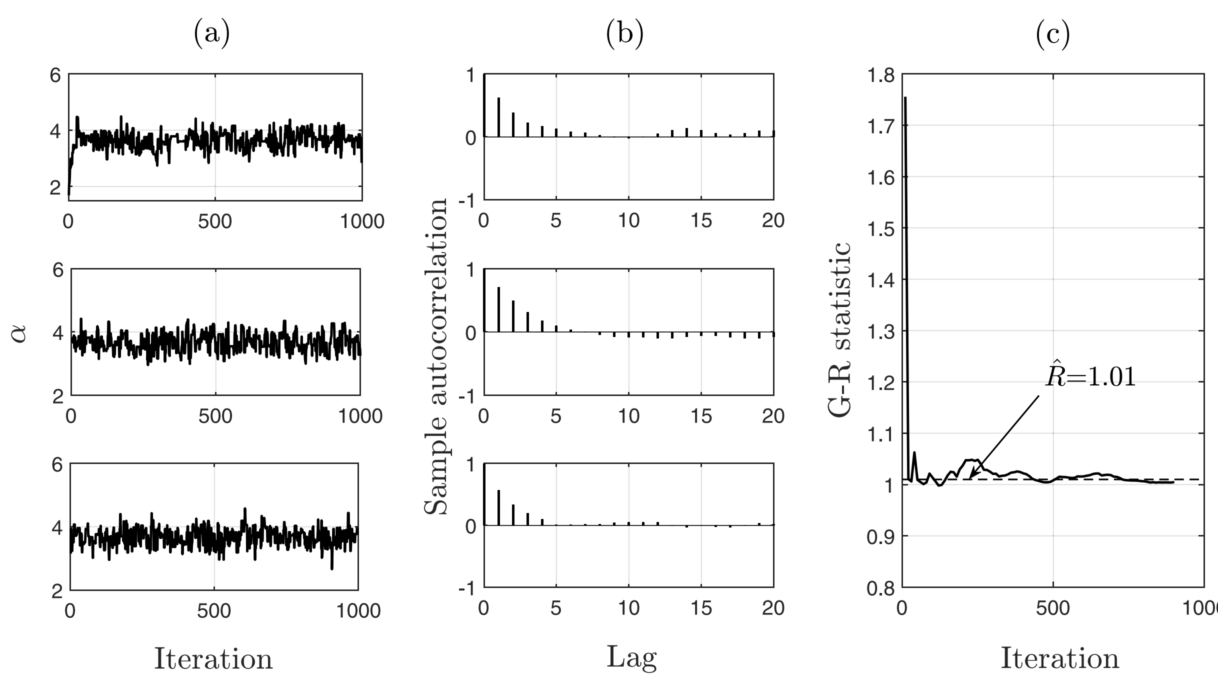

Fig. D2. (a) Three MCMC runs of the parameter α generated by L-MCMC, and the corresponding (b) sample ACF plots, and (c) the convergence of the G-R statistic (Section 4.1).

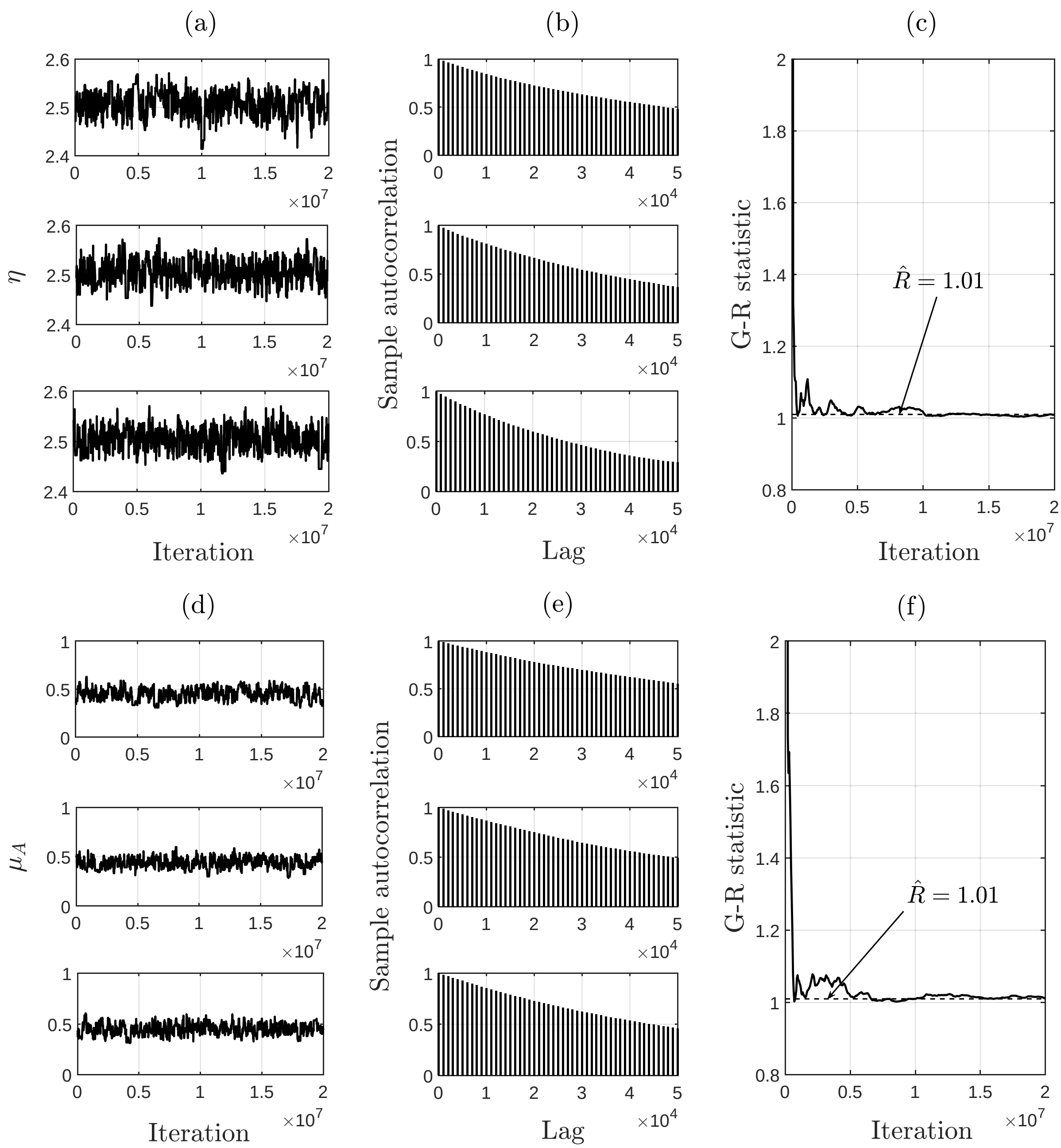

Fig. D3. Three MCMC runs of the parameters (a) η and (d) μ_A generated by ABC-MCMC with $\epsilon = 0.61$, and the corresponding (b,e) sample ACF plots, and (c,f) the convergence of the G-R statistics (Section 4.2).

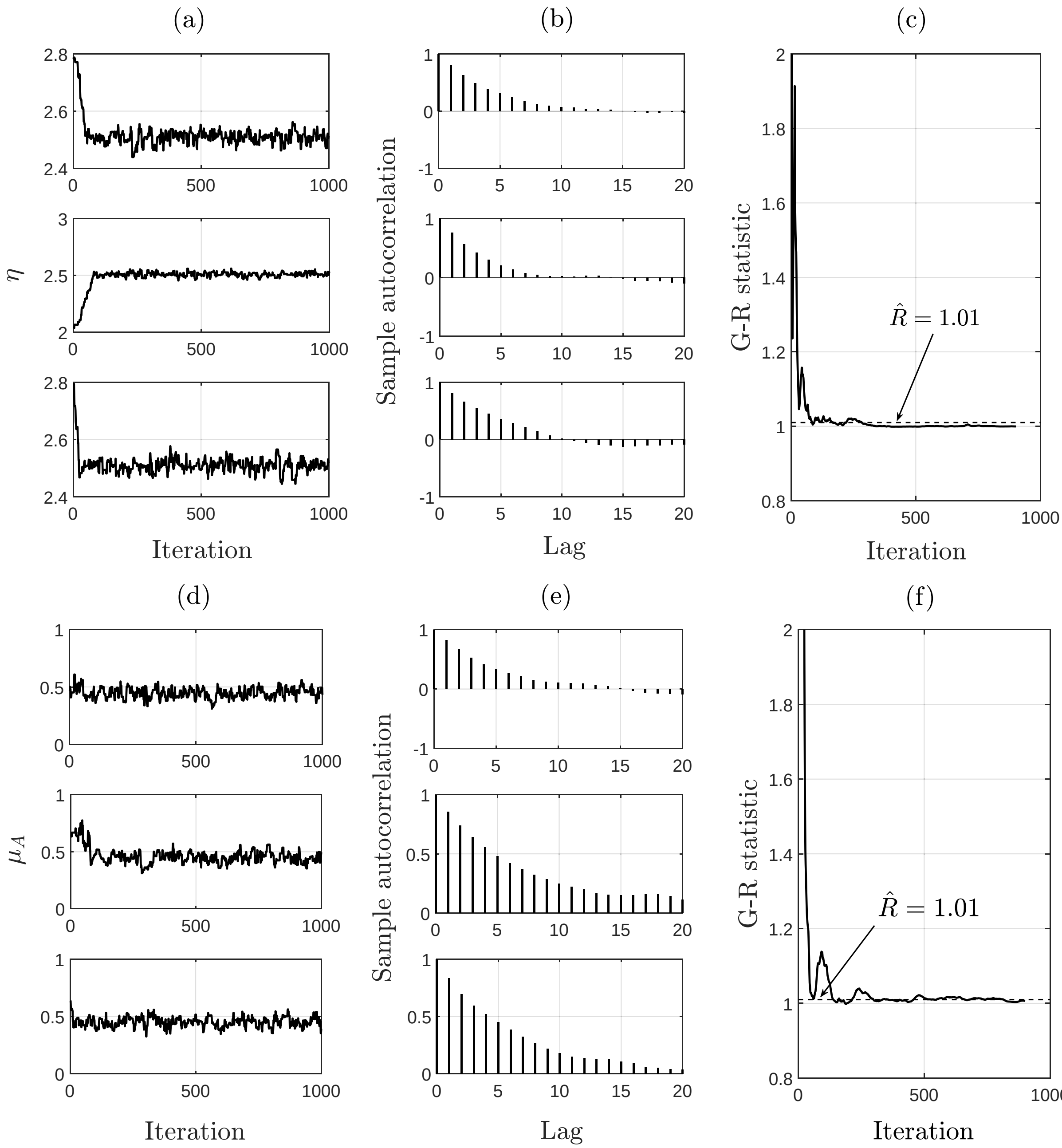

Fig. D4. Three MCMC runs of the parameters (a) η and (d) μ_A generated by L-MCMC, and the corresponding (b,e) sample ACF plots, and (c,f) the convergence of the G-R statistics (Section 4.2).

Supplementary material

Supplementary material associated with this article can be found, in the online version, at 10.1016/j.ress.2019.106780

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